Plants in limbo – the theory of detrital photosynthesis

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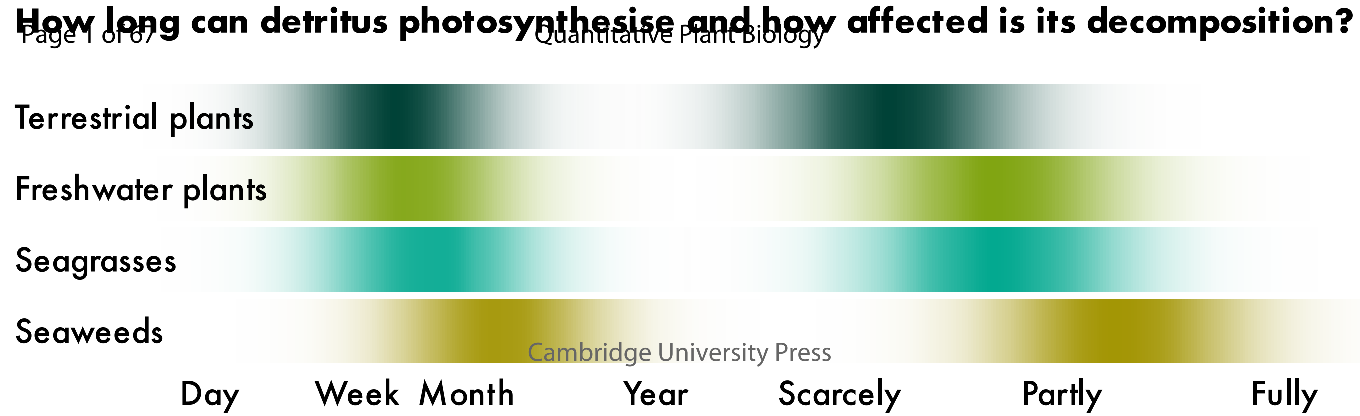
How long can detritus photosynthesise and how affected is its decomposition?
Terrestrial plants
Freshwater plants
Seagrasses
Seaweeds
Day
Week
Month
Year
Scarcely
Partly
Fully

Quantitative Plant Biology

Theories

Plants in limbo — the theory of detrital photosynthesis

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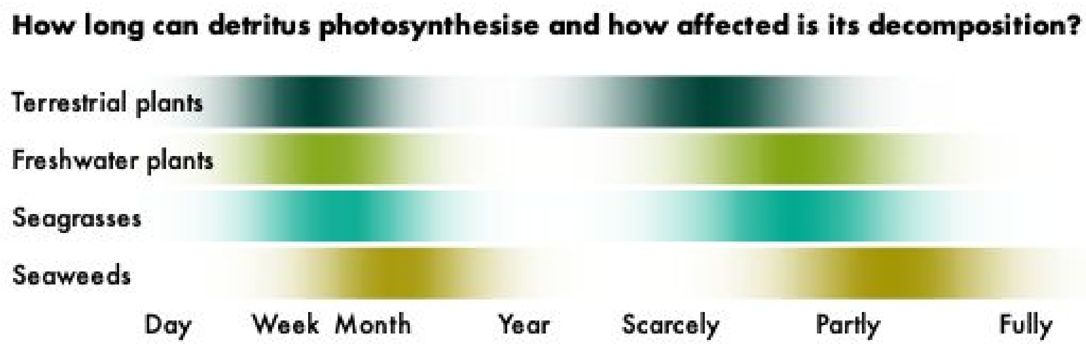
Abstract



Kelp detritus can maintain photosynthesis following detachment. This could be explained by limited tissue differentiation in non-vascular plants, but it remains unclear if detrital photosynthesis is restricted to macroalgae. Here I formalise the theory of detrital photosynthesis by developing a mathematical model and performing a meta-analysis of post-detachment physiology across 129 studies on 92 species of terrestrial and aquatic plants, including original data on the seagrasses *Amphibolis antarctica* and *Halophila ovalis*. Detrital photosynthesis is a general phenomenon, but its longevity in macroalgae is unmatched. Photosynthetic half-life increases from terrestrial plants (11 days) over freshwater Alismatales (14 days) and seagrasses (17 days) to seaweeds (44 days) and only exceeds detrital half-life for seaweeds (3:1). Detrital photosynthesis therefore uniquely determines macroalgal decomposition. This explains why light slows kelp decomposition, contrary to the expected effect of photodegradation. The presented theory questions conventional understanding of decomposition, especially in the context of blue carbon.

Keywords carbon cycle, carbon sequestration, decay, degradation, leaf litter

Introduction

…tossed and swept like dead leaves from one spot to another, never resting, never giving back their goodness to the earth, never fully dead but in some vast limbo between life and extinction, tossing and tumbling without end… — Philip Pullman (1994) *The Tin Princess*

Seaweed detrital dynamics are influenced by physiology. Kelp detritus can remain physiologically viable for months (de Bettignies et al. 2020, Frontier et al. 2021, Wright & Foggo 2021, Frontier et al. 2022, Wright et al. 2022, Wright & Kregting 2023, Wright et al. 2024), often resulting in atypical decomposition trajectories (Frontier et al. 2021, Filbee-Dexter et al. 2022, Frontier et al. 2022, Kennedy & Blain 2025). Floating kelp detritus sinks before it stops photosynthesising (Graiff et al. 2013, 2016, Tala et al. 2019). And if detached tissue happens to be reproductive, it can even be a vector for dispersal (Macaya et al. 2005, Fraser et al. 2018, Tala et al. 2019, de Bettignies et al. 2020, Frontier et al. 2021). This stands in stark contrast to the traditional ecological definition of detritus as "dead organic matter" (Moore et al. 2004, discussed in Wright & Kregting 2023). There is no word for plant material in limbo—detached but alive. Detached seaweed, except for that of holopelagic origin, has always been referred to as detritus. I propose to continue using this term *sensu lato* for any detached plant tissue and detritus *sensu stricto* for detached *and* dead plant tissue. Detrital photosynthesis—defined as the persistence of photosynthesis after detachment (Frontier et al. 2021, Wright et al. 2022, 2024)—provides the strongest evidence in support of the viability of detritus. But the fundamental impact on detrital dynamics only becomes apparent when comparing decomposition rates between live and dead or senescent detritus. Seaweed detritus with impaired physiology clearly decomposes faster (Birch et al. 1983, Brouwer 1996, Hamersley et al. 2015, de Bettignies et al. 2020). This is likely due to maintenance of growth and chemical defence (Tala et al. 2019, de Bettignies et al. 2020, Wright et al. 2022), probably making physiology an important source of detrital recalcitrance (Wright et al. 2022, 2024).

There is currently no baseline for detrital photosynthesis. The term was first introduced quite recently in the context of kelp decomposition (Frontier et al. 2021, Wright et al. 2022, 2024). But there was clearly awareness of the phenomenon among early phycologists who noted the ability of excised tissue to maintain photosynthesis (Drew 1983) and deemed it necessary to pre-kill detritus (Josselyn & Mathieson 1980, Rice & Tenore 1981, Birch et al. 1983, Smith & Foreman 1984, Brouwer 1996). Other fields of botany do not use the term but the amount of research into physiology of excised leaves of terrestrial and freshwater plants is enormous (e.g. Thimann & Satler 1979, Jana & Choudhuri 1980, 1982, Kar & Choudhuri 1986, 1987, Powles et al. 2006, Gauthier & Jacobs 2018, Kar et al. 2021). The literature also includes three studies on seagrasses (Knauer & Ayers 1977, Pellikaan 1982, Strother & Vatta 1986), the only fully marine angiosperms and vascular counterpart to seaweeds (Olsen et al. 2016). Yet no attempt has been made to align these seemingly incongruous research trajectories in a formal comparison. It consequently remains unclear if detrital photosynthesis is unique to seaweeds. It has been suggested that detrital photosynthesis may be explained by the ability of all macroalgal tissues to independently photosynthesise to some extent (Wright & Kregting 2023) but there is no certainty due to a lack of reference points. The mechanism underlying the ability of seaweeds to maintain detrital photosynthesis remains unclear. It could be their aquatic nature which removes reliance on roots for water, nutrient provisioning from the surrounding water, non-vascularity, minimal tissue differentiation and/or something altogether different. To understand the prerequisite for detrital photosynthesis and thus predict prevalence among plants, model systems beyond seaweeds are needed.

Seagrasses are suitable candidates for further investigation of detrital photosynthesis. While also marine macrophytes, seagrasses are fundamentally different from seaweeds. Seagrasses are vascular plants which, due to their terrestrial ancestry (Olsen et al. 2016), are differentiated into roots, rhizomes or stems, and leaves. However, convergent evolution with seaweeds (Olsen et al. 2016) has reduced their reliance on roots for nutrient uptake. Seagrass leaves tend to outperform their roots in suppling nutrients to the plant (Pedersen & Borum 1992, Stapel et al. 1996, Pedersen et al. 1997, Gras et al. 2003, Viana et al. 2019). This suggests that seagrass leaves can independently photosynthesise, seemingly reducing the primary function of the roots to anchorage, analogous to the holdfast of seaweeds. However, given the usually greater ammonium concentration in sediment pore water, roots can supply more nitrogen despite the leaves' greater uptake affinity (Short & McRoy 1984, Lee & Dunton 1999, but see Pedersen & Borum 1992) and nutrients are translocated to leaves (Short & McRoy 1984, Viana et al. 2019), even those tens of centimetres from the source (Marbà et al. 2002). Roots can also tap into nutrient pools that are unavailable to leaves, such as particulate (Evrard et al. 2005) and microbial (Patriquin & Knowles 1972, Mohr et al. 2021) nitrogen. Clearly the photosynthetic emancipation of seagrass leaves is at a halfway point between terrestrial plants and macroalgae, so seagrasses probably display detrital photosynthesis to some extent. Relatively slow and light-dependent chlorophyll degradation in seagrass detritus (Knauer & Ayers 1977, Pellikaan 1982, Strother & Vatta 1986) provides some support for this, but there are currently no data on seagrass detrital photosynthesis.

Here I aim to formalise the theory of detrital photosynthesis in the hope that it will provide a better model for understanding plant decomposition at large. To address this aim, I first conducted a brief *ex situ* decomposition experiment on seagrasses to fill the knowledge gap on seagrass detrital photosynthesis. I chose the paddle weed *Halophila ovalis* (Hydrocharitaceae) and wire weed *Amphibolis antarctica* (Cymodoceaceae) as diverse representatives because of their phylogenetic (diverged 105 Ma ago, Waycott et al. 2018) and morphological (de los Santos et al. 2012, 2016) differences. Senescence of fresh leaves was induced by cutting their petiole and leaving them to decompose on sediment, measuring net oxygen production of destructively sampled leaves at regular intervals over 34 days. On the basis of these and previous (Wright et al. 2022, 2024) observations of detrital photosynthesis, I then developed a mathematical model which describes detrital photosynthesis as a logistic decay function of detrital age. After compiling a detrital photosynthesis dataset (Wright 2025), I used this model in a Bayesian meta-analysis to infer the longevity of detrital photosynthesis for different plants and quantify the effects of light and choice of response variable (photosynthesis vs. chlorophyll). Finally, I compared the persistence of detrital photosynthesis with detrital half-life (Olson 1963) to determine relative physiological influence on decomposition for different plants. This is especially relevant to the case of seaweeds—the current focus of blue carbon (Krause-Jensen & Duarte 2016, Pessarrodona et al. 2023, Filbee-Dexter et al. 2024)—whose carbon cycle is speculated to be strongly influenced by detrital photosynthesis (Wright et al. 2022, 2024).

Methods

Model

Detached plant parts can remain in limbo for long durations, often photosynthesising near baseline levels. For this reason and for simplicity detrital photosynthesis has mostly been described as a linear function of detrital age (Frontier et al. 2021, Wright & Foggo 2021, Wright et al. 2022, Wright & Kregting 2023, Wright et al. 2024). But in two cases a logistic decay function was used: in a beta model predicting chlorophyll fluorescence (Fv/Fm) (Frontier et al. 2022) and in a binomial model predicting the probability of photosynthesis (Wright et al. 2024) with time after excision. The logistic function describes the decay of photosynthesis after detachment better than the linear function because it is bounded. Detrital photosynthesis cannot conceivably exceed baseline photosynthesis under identical conditions. An ill-defined baseline or changes in light, temperature, carbon, nutrients etc. may naturally lead to brief increases after detachment (Frontier et al. 2021, Wright & Kregting 2023), but this cannot be considered part of the underlying process of interest. Similarly, photosynthesis cannot decline beyond a set minimum, which is zero for most response variables (e.g. gross photosynthesis, chlorophyll fluorescence, chlorophyll) and a negative constant for net photosynthesis. Hence the logistic function is the natural choice.

There are various ways to parameterise the logistic function. Typically, a simple linear term with intercept and slope is inserted into the standard logistic function. But for logistic decay with time (t), the intercept is not a meaningful parameter because $f(t = 0)$ is the known baseline. The slope is more meaningful because it provides the logistic rate of decay per unit t, but ultimately the parameter of interest is the sigmoid inflection point or midpoint: t when $f(t) = 0.5$. It has the same units as t and is the only parameter that gives a directly interpretable measure of decay. As such it is analogous to half-life ($t_{½}$), t when $f(t) = 0.5$ for the exponential decay function (Olson 1963). The logistic midpoint can be calculated as intercept / slope, but this usually results in enormous uncertainty and difficulty to calculate meaningful central tendencies (Wright et al. 2024), so the midpoint is best directly estimated. The slope and midpoint parameterisation is sometimes used but these parameters are not identifiable with an intercept near a known baseline. To simplify the model while retaining the midpoint, one must either fix the slope or intercept. Only a constant intercept is sensible and since the baseline is expected near 1, a logistic intercept of 5, translating to $f(0) = 0.99$, is the natural choice.

The most complex case to model is that of net photosynthesis since it has a negative lower bound. Two scaling parameters outside the standard logistic function are required. Thus my full logistic function describing the decay of photosynthesis (p) with time after detachment (t) is

$$p(t) = \frac{\alpha + \tau}{1 + e^{\frac{5(t-\mu)}{\mu}}} - \tau \qquad (1)$$

where α is baseline photosynthesis, μ is the midpoint of photosynthetic decay and τ is baseline respiration. $\tau = 0$ when the lower bound is zero as for gross photosynthesis etc. and $\alpha = 1$ when photosynthesis is normalised relative to the baseline. The naming of parameters is inspired by the word אמת (graecised τμα) which is composed of the first, middle and last letters of the Hebrew alphabet, symbolising life (א), transition (מ) and death (ת). In the story of the golem, the word lends life but removing א changes its meaning from truth to death (מת) and takes life.

Literature review

Following the seagrass experiment (see methods in Supplement), I systematically searched the literature using keyword strings such as "(photosynthesis OR chlorophyll) AND ('induced senescence' OR detrit* OR detach* OR excis* OR cut OR isolate*) AND (week* OR day* OR hour* OR minute* OR time OR age)" in Web of Science (webofscience.com/wos) and Google Scholar (scholar.google.com). Once I had a selection of suitable papers, I used Crossref Metadata Search (search.crossref.org) to find similar papers. I accepted studies on all plants *sensu lato*, the condition being that senescence was induced by excision at an exact time and photosynthesis was repeatedly measured in a timeseries of at least 3 timepoints over any interval to assess viability of the detached tissue. Studies on abscission were not considered since senescence starts at an unknown time prior to detachment. I naturally did not include plants that always live detached, such as the holopelagic seaweeds *Sargassum fluitans* and *S. natans*. I accepted various response variables, including O_2 and CO_2 gas exchange, ^{14}C assimilation, Fv/Fm, photosystem I and II electron transport rates (Mehler and Hill reactions), RuBisCo activity or concentration, carbonic anhydrase activity and photorespiration (glycolate concentration), as well as total chlorophyll, chlorophyll *a* or a chlorophyll indicator. Microbial photosynthesis was generally assumed to be negligible since most of the mentioned response variables would likely not detect it. Including the present study, this resulted in 129 suitable studies between 1957 and 2025 (see meta-analysis references in Supplement) and 551 experiments (median = 3 per study).

Data were extracted by copying tables or digitising figures using WebPlotDigitizer v4.6 (Rohatgi 2022). In two cases I contacted authors for their raw data and in five cases the data were my own. Raw data were otherwise not available. The resulting dataset (Wright 2025) contains data on 92 species from 23 orders in four non-taxonomic groups of interest: terrestrial plants (66 species), freshwater plants (6 species), seagrasses (5 species) and seaweeds (15 species). It is noteworthy that all freshwater plants in the dataset belong to the order Alismatales, like seagrasses, and all seaweeds belonged to the phyla Chlorophyta and Heterokontophyta (Table S1). The assigned taxonomy follows Plants of the World Online (POWO 2025) and AlgaeBase (Guiry & Guiry 2025). My distinction between freshwater and terrestrial plants is based on whether leaves are submerged and photosynthesis can therefore be considered aquatic. There were no studies on freshwater macroalgae, so seaweeds are assumed to be representative. Several predictor variables including light- and water availability as well as the type of photosynthesis measurement were also recorded. When individual observations could not be extracted, the mean, standard error of the mean (s.e.m.) and sample size (n) were recorded instead. Uncertainty was mostly given as s.e.m., but when standard deviation (s.d.) was provided instead, it was converted as $\text{s.e.m.} = \frac{\text{s.d.}}{\sqrt{\text{n}}}$. In a few cases uncertainty was given as a 95% confidence interval (CI) or interquartile range (IQR). In the former case, I converted as $\text{s.e.m.} = \frac{0.5 \times \text{CI}_{0.95}}{\Phi^{-1}(0.975)}$, in the latter I assumed mean = median and conservatively converted the larger of the two quartile ranges ($Q_3 - Q_2$ or $Q_2 - Q_1$) as $\text{s.e.m.} = \frac{\text{QR}}{\Phi^{-1}(0.75) \times \sqrt{\text{n}}}$.

Meta-analysis

As is typical of meta-analyses, the data contain a mixture of observations and means. To jointly analyse data with at different levels of uncertainty one must either summarise observations and build a measurement error model or expand means to observations by simulation. I opted for the latter and obtained 10646 observations by simulating draws from the sampling distribution. For most response variables, which are bounded by zero, I chose the gamma distribution, for net photosynthesis I chose the normal distribution and for Fv/Fm I chose a truncated normal distribution because some summary statistics violated the beta distribution. I implemented the sampling in R v4.5.2 (R Core Team 2026) using `rgamma( n , mean^2 / sd^2 , mean / sd^2 )`, `rnorm( n , mean , sd )` and `rtnorm( n , mean , sd , 0 , 1 )`, the last function being from extraDistr v1.10.0 (Wolodzko 2023).

Since the response variable is an amalgam of from various measures of photosynthesis, I normalised it to enable formal comparison. Observations were expressed relative to the initial observation or mean of initial observations in the timeseries. The resulting ratios, representing relative photosynthesis, are the least invasive form of normalisation and frequently used to express photosynthetic decay (29 of 129 studies, e.g. Thimann & Satler 1979, Jana & Choudhuri 1980, Pellikaan 1982). The drawback of this approach is that data are not fully brought onto the same scale because net photosynthesis allows zeros and negative ratios. I resolved this issue by replacing all zeros and negative ratios with the minimum observed positive ratio, assumed to be the minimum observable ratio within measurement error of zero. This means equating net respiration with absence of photosynthesis, which is of course not the case because respiration can mask photosynthesis in net photosynthesis measurements. Nonetheless, I believe this to be acceptable for the purposes of determining the timescales of photosynthetic decay. This normalisation allows using the simplest form of Equation 1 with $\tau = 0$ and $\alpha = 1$.

Apart from t (days), the main predictor variables of interest are plant group (terrestrial plants, freshwater plants, seagrasses, seaweeds), light availability and the type of response variable (chlorophyll or photosynthesis). Light is naturally a key predictor of photosynthetic persistence (Frontier et al. 2021, 2022, Wright et al. 2024) and the type of response variable is important because chlorophyll and photosynthesis are expected to senesce on different timescales. Water availability is only relevant to terrestrial plants and was therefore not considered. Since light availability and the type of response variable are binary predictors, they were coded as indicators of light (−0.5 for dark, +0.5 for light) and chlorophyll (−0.5 for photosynthesis, +0.5 for chlorophyll), centred on zero to avoid the bias in prior variance typical of indicator variables

(McElreath 2020) and allow easy prediction of the mean. To get a sense of interspecific and experimental variation, I also included plant species (92 levels) and experiment (551 levels) as predictors. The resulting Bayesian log-normal multilevel model of relative photosynthesis (p) is

$$\ln p \sim \mathcal{N}(\,\overline{p}\,,\sigma\,) \tag{2}$$

$$\overline{p} = \ln \frac{1}{1+e^{\frac{5(t-\mu)}{\mu}}}$$

$$\ln \mu = \alpha_G + \alpha_S + \alpha_E + \beta_G \times l + \gamma_G \times c$$

$$\alpha_G \sim \mathcal{N}(\,\overline{\alpha}\,,\sigma_{\alpha_G}\,)$$

$$\alpha_S \sim \mathcal{N}(\,0\,,\sigma_{\alpha_S}\,)$$

$$\alpha_E \sim \mathcal{N}(\,0\,,\sigma_{\alpha_E}\,)$$

$$\beta_G \sim \mathcal{N}(\,\overline{\beta}\,,\sigma_{\beta}\,)$$

$$\gamma_G \sim \mathcal{N}(\,\overline{\gamma}\,,\sigma_{\gamma}\,)$$

$$\overline{\alpha} \sim \mathcal{N}(\,\ln 14\,,0.3\,)$$

$$\overline{\beta},\overline{\gamma} \sim \mathcal{N}(\,0, 0.4\,)$$

$$\sigma_{\alpha_G},\sigma_{\alpha_S},\sigma_{\alpha_E} \sim \mathcal{N}_{0,\infty}(\,0, 0.3\,)$$

$$\sigma_{\beta},\sigma_{\gamma} \sim \mathcal{N}_{0,\infty}(\,0, 0.4\,)$$

$$\sigma \sim Exp(\,1\,)$$

where the logarithm of μ (Equation 1) is described by a linear model with intercepts (α) indexed by plant group (G), species (S) and experiment (E), the effect of light (β) and the effect of measuring chlorophyll relative to photosynthesis (γ) indexed by G. β and γ are differences on the logarithmic scale, so represent log ratios. All parameters are partially pooled to enable prediction for new categories, borrow information and regularise spurious effects (McElreath 2020). The intercept standard deviations (σ_α) give an indication of intergroup, interspecific and experimental variance. The prior for the mean of μ on the logarithmic scale ($\overline{\alpha}$) is centred on ln 14 because two weeks seems like a reasonable median midpoint of photosynthetic decay based on my observations on kelps (Wright et al. 2022, 2024) and literature review. Even though is suspected β and γ to be positive, I centred the priors for $\overline{\beta}$ and $\overline{\gamma}$ on zero to avoid biasing the effect.

The model was written in Stan (Carpenter et al. 2017) and implemented with CmdStan v2.17.1 (Lee et al. 2017) in R. This was facilitated by cmdstanr v0.9.0 (Gabry et al. 2025), tidybayes v3.0.7 (Kay 2024) and bayesplot v1.14.0 (Gabry et al. 2019). Posterior probability distributions for each parameter were estimated based on 8×10^4 Markov chain Monte Carlo samples (10^4 iterations × 8 chains). Model performance was assessed using $\hat{R}$ scores and trace rank and pair plots (Gabry et al. 2019). For data wrangling, summary and visualisation I used tidyverse v2.0.0 (Wickham et al. 2019), here v1.0.2 (Müller 2025), ggridges v0.5.7 (Wilke 2021), ggh4x v0.3.1 (van den Brand 2025), geomtextpath v0.2.0 (Cameron & van den Brand 2022), patchwork v1.3.2 (Pedersen 2025) and officer v0.7.0 (Gohel et al. 2025). For details, see data and code at github.com/lukaseamus/plant-limbo.

To contextualise μ (days), I formally compared it with the half-life $t_{½}$ (days), its equivalent in the exponential model of decomposition. To facilitate the comparison, I hereinafter refer to μ as the photosynthetic half-life and $t_{½}$ as the detrital half-life. $t_{½}$ is related to the exponential decay constant k by $t_{½} = \ln 2 / k$ (Olson 1963). k is usually estimated using the exponential decay function but can also be approximated by the detrital turnover rate, the ratio of detrital production to the detrital pool (Cebrián & Lartigue 2004). Using published data (Cebrián & Lartigue 2004, Anderson-Teixeira et al. 2018), I thus extracted or calculated k for terrestrial plants, freshwater plants, seagrasses and seaweeds, and converted it to $t_{½}$. I then modelled $t_{½}$ with a log-normal multilevel intercept model and compared its predictions to those for μ from Equation 2. I formalised this comparison as the ratio $\mu / t_{½}$, a measure of physiological influence on decomposition. $\mu / t_{½} = 1$ implies that detrital photosynthesis reaches half its initial value at the same time as detrital mass. If $\mu / t_{½} < 1$, photosynthesis decays faster than mass and is unlikely to influence decomposition but if $\mu / t_{½} > 1$, photosynthesis essentially determines decomposition.

Results

Seagrasses

Halophila ovalis photosynthesised for longer post-excision than *Amphibolis antarctica*. After 34 days, the duration of the experiment, all excised *H. ovalis* leaves were still obviously net autotrophic while all excised *A. antarctica* leaves were apparently dead. On a mass basis, *H.*

ovalis displayed similar baseline light-saturated net photosynthesis (P_{max}) to *A. antarctica*, but its photosynthetic half-life (μ in Equation 1) was 3.9 ± 1.6 (mean ± standard deviation) times as long (Figure 1, Table 1). On a leaf basis, baseline P_{max} of *A. antarctica* was naturally much greater but again the photosynthetic half-life of *H. ovalis* was 6.8 ± 2.5 times as long (Table 1).

Mean leaf blotted mass of *H. ovalis* (initially 0.052 ± 0.005 g leaf^{-1}) declined at linear rates of 0.58 ± 0.26 mg leaf^{-1} d^{-1} or 1.1 ± 0.43 % d^{-1} while that of *A. antarctica* (initially 0.64 ± 0.053 g leaf^{-1}) only declined at 2 ± 2.7 mg leaf^{-1} d^{-1} or 0.3 ± 0.41 % d^{-1}. Hence it is surprising that excised *H. ovalis* leaves photosynthesised for so much longer. The most likely reason is that periphyton, which I observed colonising *H. ovalis* but not *A. antarctica*, contributed to photosynthesis. This likely exacerbated the observed difference in detrital viability between the two seagrasses. Accounting for these limitations, I expect the average seagrass to have a photosynthetic half-life of more than a week but less than a month.

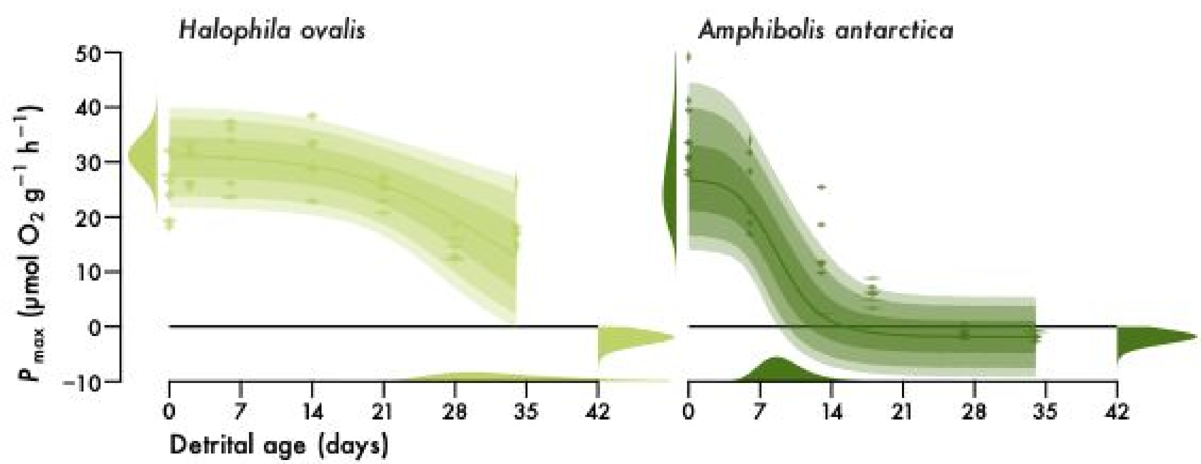


Figure 1. Seagrass detrital photosynthesis. Light-saturated net photosynthesis (P_{max}) of excised *Halophila ovalis* and *Amphibolis antarctica* leaves over 34 days post-excision. P_{max} is standardised by leaf fresh mass. Violins are kernel density estimates of observations with measurement error (n = 42 for *H. ovalis*, n = 39 for *A. antarctica*). Lines and intervals are medians and the central 50%, 80% and 90% of posterior probability for predicted observations. Distributions are posteriors for initial P_{max} (α), photosynthetic half-life (μ) and final respiration (τ) correctly mapped onto the axes (cf. Equation 1).

Table 1. Seagrass parameter estimates (mean ± standard deviation). Δ is the absolute difference between estimates for *H. ovalis* and *A. antarctica*. $P(\Delta > 0)$ is the probability that estimates for *H. ovalis* and *A. antarctica* are different. $P = 0.5$ is equivalent to a coin toss.

Parameter	*Halophila ovalis*	*Amphibolis antarctica*	Δ	$P(\Delta > 0)$
Mass-based				
α (µmol g^{-1} h^{-1})	31 ± 3.4	28 ± 8.4	3.3 ± 9.1	0.69
μ (days)	35 ± 11	9.5 ± 2.6	26 ± 11	1
τ (µmol g^{-1} h^{-1})	2.1 ± 0.85	1.9 ± 0.76	0.17 ± 1.1	0.56
Leaf-based				
α (µmol leaf^{-1} h^{-1})	1.6 ± 0.17	33 ± 12	31 ± 12	1
μ (days)	46 ± 15	7 ± 1.3	39 ± 15	1
τ (µmol leaf^{-1} h^{-1})	0.92 ± 0.38	0.93 ± 0.37	0.0049 ± 0.53	0.51

Meta-analysis

Comparison of detrital photosynthesis across various plants revealed that light ameliorates photosynthetic decay. Detritus of terrestrial plants, seagrasses and seaweeds with access to light is predicted to have a photosynthetic half-life that is 1.8, 1.6 and 2 times as long as that in darkness (Figure 2, Table 2). Freshwater plants form an exception with a mere 68% chance (only somewhat above 50:50) of extended photosynthesis under light (Figure 2, Table 2). The predicted median ratio for new plants is therefore 1.5 (ln ratio = 0.4 ± 0.52). In absolute terms, however, the light effect is only substantial for seaweeds (Figure 2). While median terrestrial photosynthetic half-life is only expected to increase from 7.2 to 13 days with access to light, that of seaweeds increases from 24 to 51 days (Table S2). In darkness, seaweed detritus thus only remains viable for comparable durations to other plants (Figure 2, Table S2). This suggests that the detrital longevity of seaweeds is highly light-dependent.

The choice of response variable also matters when estimating detrital photosynthesis. Chlorophyll generally decays slower than photosynthesis. This pattern is consistent across all plants with photosynthetic half-life predicted to be 1.4 times (ln ratio = 0.33 ± 0.36) as long for chlorophyll as for photosynthesis (Figure 2, Table 2). In absolute terms, the effect again naturally dominates for seaweeds (Figure 2), with an increase from 51 days for photosynthesis to 81 days for chlorophyll under light (Table S2). Measurements based on chlorophyll therefore tend to overestimate the longevity of detrital photosynthesis, especially for seaweeds.

Across light conditions and response variables, the median half-life of detrital photosynthesis ranges from 11 days for terrestrial plants, over 14 and 17 days for freshwater plants and seagrasses to 44 days for seaweeds (Table 2). The prediction for new plants is 17 days (ln days = 2.8 ± 1.1). In relative terms, seaweed detrital photosynthesis persists 3.9, 3.1, and 2.5 times as long as that of terrestrial plants, freshwater plants and seagrasses (Table 3, Figure S2). All vascular plant detritus remains viable for similar durations but there is a tendency for that of freshwater plants to have a somewhat longer photosynthetic half-life compared to terrestrial plants, with that of seagrasses being longer still (Table 3, Figure S2). This suggests that the non-vascular macroalgae have the greatest potential for detrital photosynthesis while among the vascular plants there is increased detrital viability with a shift from a terrestrial to an aquatic and yet again to a marine lifestyle.

Table 2. Meta-analysis parameter estimates (Equation 2) for plant groups. Photosynthetic half-life (μ) is predicted for new experiments on unobserved species across the light (β) and chlorophyll (γ) effects. Exponentiating β and γ yields relative effect sizes (cf. Figure 2). μ is also expressed relative to detrital half-life ($t_{½}$) to quantify the physiological influence on decomposition (cf. Figure 3). All estimates except P are given as median (ln mean ± standard deviation).

Parameter	Terrestrial plants	Freshwater plants	Seagrasses	Seaweeds
μ (days)	11 (2.4 ± 0.92)	14 (2.7 ± 0.95)	17 (2.9 ± 0.97)	44 (3.8 ± 0.96)
e^{β}	1.8 (0.61 ± 0.18)	1.2 (0.16 ± 0.37)	1.6 (0.47 ± 0.4)	2 (0.77 ± 0.47)
$P(\beta > 0)$	1	0.68	0.89	0.98
e^{γ}	1.3 (0.29 ± 0.15)	1.5 (0.41 ± 0.26)	1.4 (0.32 ± 0.3)	1.5 (0.47 ± 0.29)
$P(\gamma > 0)$	0.97	0.96	0.89	0.98
$t_{½}$ (days)	409 (6 ± 1.5)	57 (4 ± 1.5)	68 (4.2 ± 1.5)	14 (2.7 ± 1.5)
$\mu/t_{½}$	0.028 (−3.6 ± 1.8)	0.25 (−1.4 ± 1.8)	0.25 (−1.4 ± 1.8)	3 (1.1 ± 1.8)
$P(\mu/t_{½} > 1)$	0.02	0.22	0.23	0.73

Notwithstanding these general trends, there is substantial interspecific and experimental variation in photosynthetic half-life (Figure 3). Median estimates for species range from 3–4 days for the wheat *Triticum aestivum* to 154 days for the Antarctic brown seaweed *Desmarestia anceps*

(Table S3). Median estimates for experiments range from 1.2 to 438 days. The variance in photosynthetic half-life between plant groups, species and experiments is directly quantified by logarithmic partial pooling standard deviations (σ_α in Equation 2). These inter-group, interspecific and experimental standard deviations are 0.51 ± 0.16, 0.53 ± 0.12 and 0.72 ± 0.057, so experimental variation is the greatest source of uncertainty. Such variation in the physiological viability of detritus may translate to experimental variation in decomposition trajectories.

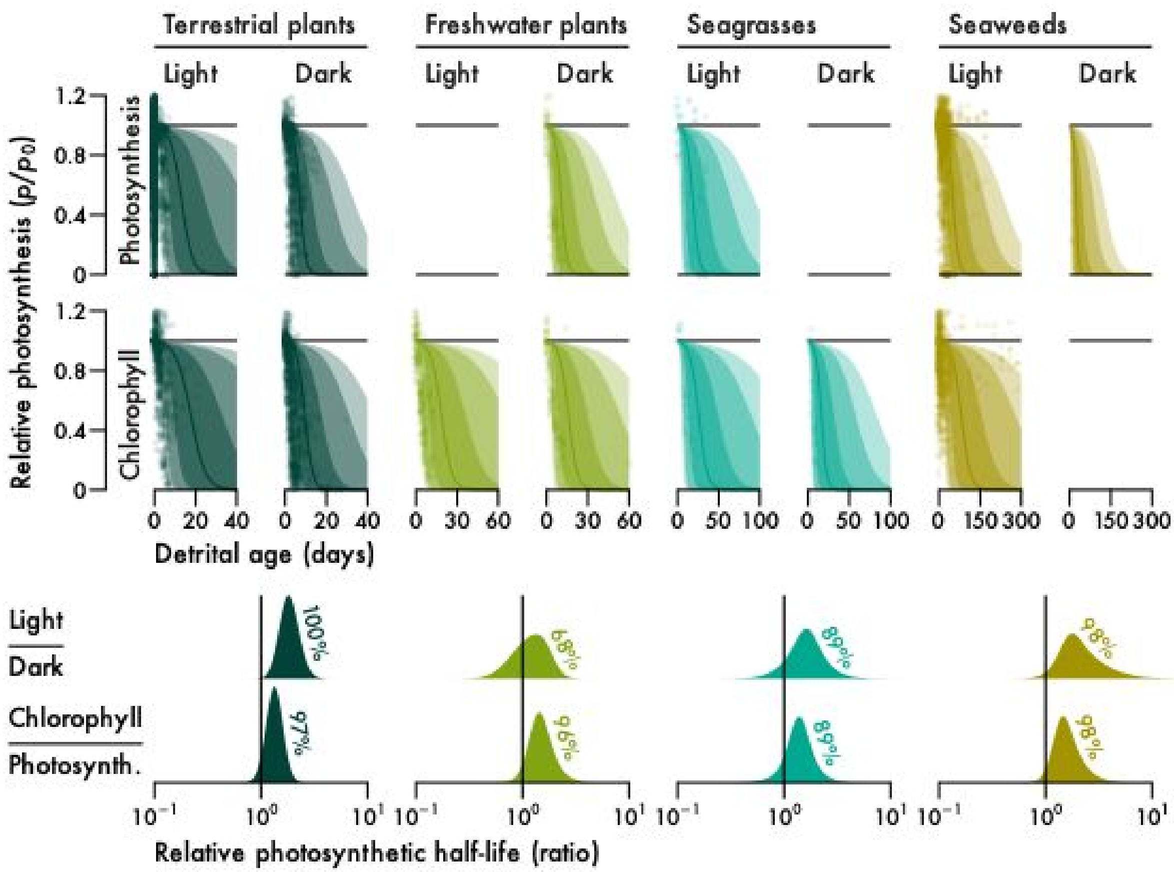


Figure 2. The effect of light and response variable type on detrital photosynthesis for major plant groups: terrestrial plants (Streptophyta excluding aquatic plants), freshwater plants (Alismatales excluding seagrasses), seagrasses, and seaweeds (Chlorophyta and Heterokontophyta excluding freshwater macroalgae). Points are observations (n = 7240, 673, 160 and 2573 for terrestrial plants, freshwater plants, seagrasses and seaweeds). Note that there are several observations outside the axes ranges. Lines and intervals are medians and the central 50%, 80% and 90% of

posterior probability for the median prediction ($e^{\bar{p}}$ in Equation 2). Distributions are kernel density estimates of posterior probabilities for the light and chlorophyll relative effect sizes (e^{β} and e^{γ} in Equation 2). Percentages are probabilities for $\beta > 0$ and $\gamma > 0$.

The extent to which detrital photosynthesis can influence decomposition not only depends on photosynthetic half-life because decomposition timescales also vary substantially across plant groups. Terrestrial plant detritus decomposes very slowly, only reaching half its initial mass after over a year on average (Figure 3, Table 2). Aquatic angiosperms decompose much faster, with a median detrital half-life ranging from 57 days for freshwater plants to 68 days for seagrasses (Figure 3, Table 2), only 14% to 17% of terrestrial half-life (Table 3). Seaweeds decompose fastest, with a median detrital half-life of only two weeks (Figure 3, Table 2), 21–26% of aquatic angiosperm half-life and only 3.5% of terrestrial half-life (Table 3). Photosynthetic half-life must therefore be viewed in relation to detrital half-life to assess how affected decomposition is by detrital photosynthesis. The ratio of photosynthetic to detrital half-life quantifies this relative influence, where ratios below 1 indicate little influence and ratios above 1 indicate substantial influence.

Macroalgal decomposition is uniquely affected by physiology. The median ratio of photosynthetic to detrital half-life ranges from 0.028 for terrestrial plants over 0.25 for freshwater plants and seagrasses to 3 for seaweeds (Figure 3, Table 2). Seaweeds are therefore the only plants with a median ratio above 1 and there is a 73% chance that the ratio for any seaweed is above 1, while the probability for aquatic angiosperms is 22–23% and only 2% for terrestrial plants (Figure 3, Table 2). This means that although all plants display detrital photosynthesis to some extent, it does not influence the decomposition of all plants. Terrestrial decomposition is almost certainly not influenced by physiology, while that of aquatic vascular plants is partly affected and macroalgal decomposition is physiologically determined.

Table 3. Photosynthetic (μ) and detrital ($t_{½}$) half-life contrasts between plant groups (cf. Figure S2). *P*(ratio > 1) is the probability that estimates are different. *P* = 0.5 is equivalent to a coin toss. All estimates except *P* are given as median (ln mean ± standard deviation).

Ratio	μ/μ	$P(\mu/\mu > 1)$	$t_{½}/t_{½}$	$P(t_{½}/t_{½} > 1)$
Seaweeds / Terrestrial	3.9 (1.4 ± 1.3)	0.85	0.035 (−3.3 ± 2.1)	0.057
Seaweeds / Freshwater	3.1 (1.1 ± 1.4)	0.80	0.26 (−1.4 ± 2.1)	0.26
Seaweeds / Seagrasses	2.5 (0.93 ± 1.4)	0.75	0.21 (−1.6 ± 2.1)	0.23
Seagrasses / Terrestrial	1.5 (0.43 ± 1.3)	0.63	0.17 (−1.8 ± 2.1)	0.2
Freshwater / Terrestrial	1.3 (0.23 ± 1.3)	0.57	0.14 (−2 ± 2.1)	0.18
Seagrasses / Freshwater	1.2 (0.2 ± 1.3)	0.56	1.2 (0.18 ± 2.2)	0.53

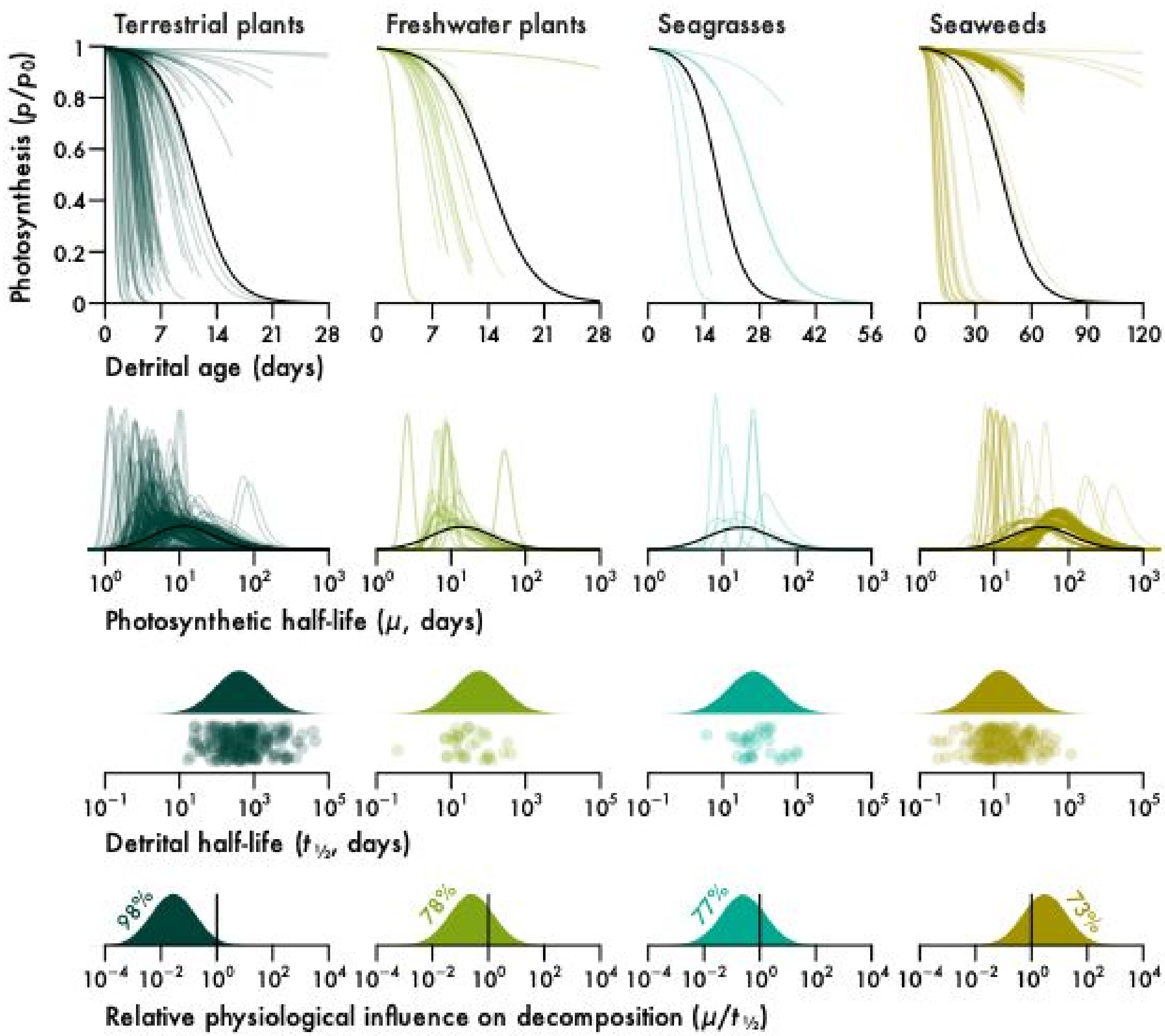


Figure 3. Longevity of detrital photosynthesis relative to detrital mass for major plant groups (cf. Figure 2). Lines are median predictions ($e^{\bar{p}}$ in Equation 2) or posterior probability distributions of photosynthetic half-life (μ in Equation 2). Existing experiments (n = 314, 32, 8 and 197 for terrestrial plants, freshwater plants, seagrasses and seaweeds) are given in colour and predictions for new experiments on unobserved species across the light and chlorophyll effects are given in

black. Raincloud plots of detrital half-life ($t_{½}$ *sensu* (Olson 1963) show observations (n = 239, 33, 32 and 171 for terrestrial plants, freshwater plants, seagrasses and seaweeds) below posterior probability distributions for new observations. Distributions in the bottom panel are ratios of the black distributions of μ to the distributions of $t_{½}$. Percentages are probabilities for $\mu/t_{½} < 1$ or $\mu/t_{½} > 1$, depending on what side of the identity line most of the probability lies.

Discussion

Some plant detritus can maintain photosynthesis following detachment. Despite the existence of various relevant literature published on this topic since the 1950s, the phenomenon of detrital photosynthesis had not been explored beyond kelps. In part, this is probably due to diverse research objectives and terminology which obscure the connection between quantitatively comparable data. There was also data scarcity for seagrasses which, as the only vascular counterpart to seaweeds, form an important functional link to freshwater and terrestrial plants. Here I formalised the theory of detrital photosynthesis by collecting the first data on photosynthesis in seagrass detritus, developing a mathematical model of detrital photosynthesis and using it to make the first quantitative comparison of photosynthetic persistence in the detrital phase of various plants. In relation to decomposition timescales estimated by exponential decay, the longevity of detrital photosynthesis provides insight into physiological influence on the fate of detritus. I predict that detrital photosynthesis is a general phenomenon but its effect on decomposition ranges from negligible for terrestrial plants over measurable for aquatic vascular plants to substantial for macroalgae. In fact, it is likely that physiology determines macroalgal decomposition, which would make the initial physiological state of detritus and its light environment the most important predictors of decomposition.

Detrital photosynthesis *per se* is not restricted to non-vascular plants. Seagrass detritus can also photosynthesise for at least a couple weeks. The physiology of diverse seagrasses responds similarly to detrital age and observed differences may be attributable to varying colonisation by periphyton which, given a leaf mass per area of 2.7 and 6.8 mg cm^{-2} for *Halophila ovalis* and *Amphibolis antarctica* respectively (de los Santos et al. 2012, 2016), could contribute 12–172% of initial seagrass photosynthesis (Neely & Wetzel 1997). Alternatively, *A. antarctica* leaves may be more reliant on their roots for nutrient provisioning. But there is no evidence to suggest this (Pedersen et al. 1997) and the 26–39-day shorter photosynthetic half-life of isolated *A. antarctica* leaves compared to *H. ovalis* is surprising given their 1.7 times ($\Delta = 47$ days) longer attached lifespan (Hemminga et al. 1999). Seagrass detrital photosynthesis, despite persisting for weeks, may not necessarily counteract decomposition. Seagrass detritus has a half-life of 68 days on average and complete decomposition may take a year (Harrison 1989, Cebrián et al. 1997, Cebrián & Lartigue 2004, Trevathan-Tackett et al. 2020). While live *Halophila decipiens* ramets do indeed decompose slower than buried ones (Kenworthy et al. 1989), decomposition rates are similar between live and senescent or buried *Thalassia testudinum* leaves (Rublee & Roman 1982, Fourqurean & Schrlau 2003, Rosch & Koch 2009) and reviews suggest that live seagrass leaves actually tend to decompose faster than senescent or dead ones (Harrison 1989, Trevathan-Tackett et al. 2020).

Macroalgae uniquely exhibit long-term detrital photosynthesis, with extreme cases persisting up to a year. As expected from their aquatic lifestyle, seagrasses and freshwater plants exhibit intermediate detrital longevity between seaweeds and terrestrial plants, whose leaves photosynthesise for less than two weeks following excision. The extreme contrast to terrestrial plants has been demonstrated but is in fact somewhat dampened by my formal analysis. With access to light but not water, which is the normal fate of terrestrial plant detritus, detached leaves of various trees tend to lose the ability to photosynthesise in just a few minutes (e.g. (Gauthier & Jacobs 2018, Kar et al. 2021) due to breakdown of stomatal conductance following a loss in water pressure (Powles et al. 2006, Gauthier & Jacobs 2018). Surprisingly seagrasses detritus tends to photosynthesise for longer than that of freshwater Alismatales. The adoption of a marine lifestyle and convergent evolution with seaweeds (Olsen et al. 2016) may be the ultimate reason. But a more proximate explanation is excess production of hydrogen peroxide in detached freshwater plant leaves (Jana & Choudhuri 1982, Kar & Choudhuri 1986, 1987). As demonstrated for *Hydrilla verticillata* (Kar & Choudhuri 1986, 1987), hydrogen peroxide production is exacerbated by photosynthesis, leading to light-induced chlorophyll degradation that is independent of photodegradation. Hence freshwater plants were the only group for which I did not find a positive light effect on photosynthetic half-life. In contrast, there is no evidence for increased chlorophyll breakdown in isolated seagrass leaves exposed to light (Strother & Vatta 1986) with photodegradation only being apparent in dead leaves (Vähätalo et al. 1998).

Perhaps the answer lies in alterations to the light harvesting complexes and cell wall features of seagrass leaves that enhance gas exchange (Olsen et al. 2016). Regardless, emancipation of leaf photosynthesis from the rest of the plant is apparently favoured by an aquatic but even more so by a marine lifestyle.

Macroalgal decomposition is uniquely determined by physiology. The influence of detrital photosynthesis on decomposition depends on its longevity relative to decomposition timescales. Of all considered plants, terrestrial ones decompose slowest and macroalgae decompose fastest, so the observed divergence in photosynthetic decay is increased when viewing it relative to decomposition. Macroalgae are the only plants whose detritus photosynthesises for as long as it remains intact, aquatic vascular plants maintain photosynthesis for a quarter of their detrital phase and terrestrial decomposition is unaffected by physiology due to a mismatch between rapid photosynthetic decay and slow decomposition. Since light increases photosynthetic half-life, detrital photosynthesis provides a mechanism contrary to photodegradation by which light slows macroalgal decomposition. Whether these predictions based on excision-induced senescence are realistic depends on the initial physiological state of detritus. Seagrass detritus often incudes green leaves (Knauer & Ayers 1977, Ochieng & Erftemeijer 1999, Jiménez-Ramos et al. 2023). This may in part be due to low nutrient resorption prior to abscission relative to terrestrial plants (Harrison 1989, Pedersen & Borum 1992, Pedersen et al. 1997, Hemminga et al. 1999, Rosch & Koch 2009). But the most likely mechanism is premature detachment caused by external factors such as hydrodynamics and herbivory (Cebrián et al. 1997) paired with seasonal weakening of the ligule (Jiménez-Ramos et al. 2023). Seaweed detritus includes whole plants which are familiar as stranded wrack but many of which are not cast up on the beach to die (van der Mheen et al. 2024, Simpkins et al. 2025). Dislodged plants account for 26% (median, range = 5.6–74%) of kelp detrital production (Gerard 1976, Newell et al. 1982, de Bettignies et al. 2013, Pessarrodona et al. 2018, Pedersen et al. 2020, Gilson et al. 2023) and are not expected to exhibit impaired photosynthesis like distally eroded or abscised tissues (Küppers & Kremer 1978, Arnold & Manley 1985, Rodriguez et al. 2016). Some viable seagrass and seaweed detritus is clearly produced in nature and will therefore follow the predictions of my model.

The theory of detrital photosynthesis is at odds with conventional understanding of the macroalgal carbon cycle. Seaweed blue carbon has been proposed as a substantial contributor to marine carbon sequestration (Krause-Jensen & Duarte 2016, Pessarrodona et al. 2023, Filbee-Dexter et al. 2024). Each proposed sequestration pathway (Pessarrodona et al. 2023) naturally hinges on decomposition. Even sequestration estimates based on deep ocean retention which does not discriminate between organic and inorganic carbon (Filbee-Dexter et al. 2024) are affected by decomposition (White & Norkko 2025). I have previously suggested that detrital photosynthesis is an important determinant of kelp decomposition (Wright et al. 2022, 2024) and have here shown this effect to be unique to macroalgae. The key implication is that seaweed decomposition is not comparable to that of other plants. Due to enormous physiological variation in seaweed detritus, as shown here, it is even hard to compare decomposition within seaweeds when not explicitly accounting for detrital photosynthesis (Kennedy & Blain 2025, White & Norkko 2025). Most of our understanding of seaweed decomposition is based on freshly excised blades (Frontier et al. 2021, Filbee-Dexter et al. 2022, Frontier et al. 2022, Wright et al. 2022, Kennedy & Blain 2025, White & Norkko 2025) as opposed to senescent fragments (Hamersley et al. 2015, de Bettignies et al. 2020) or dead tissue (Josselyn & Mathieson 1980, Rice & Tenore 1981, Birch et al. 1983, Smith & Foreman 1984, Brouwer 1996). But as mentioned, only 26% of seaweed detritus can be considered fully viable when exported. Additionally, decomposition experiments are usually carried out in mesocosms or the shallow subtidal with access to light, but seaweed carbon sinks are said to be found in the darkness of the deep sea (Krause-Jensen & Duarte 2016, Filbee-Dexter et al. 2024, van der Mheen et al. 2024, Simpkins et al. 2025). My meta-analysis suggests that darkness has a detrimental effect on the detrital photosynthesis of seaweeds, thus accelerating their decomposition. Perhaps it is time to question existing preconceptions in macroalgal blue carbon.

To conclude, I show that all plants may photosynthesise following detachment, but timescales vary greatly from a few minutes in tree leaves without access to water to a year for certain seaweeds. Only macroalgal detrital photosynthesis decays on similar timescales to detrital mass. Physiology is therefore expected to determine the decomposition of macroalgae, a relationship which is unique among plants. Predictors of photosynthesis, such as temperature and light, can hence be used to predict macroalgal decomposition. With the theory of detrital photosynthesis formalised, we now need (1) a model of macroalgal decomposition that takes into account this theory, (2) seagrass and seaweed decomposition experiments comparing live and dead detritus to directly investigate the proposed effect of physiology on decomposition (cf. Birch et al. 1983, Brouwer 1996, Hamersley et al. 2015, de Bettignies et al. 2020), (3) studies on the initial physiological state of detritus produced by erosion, abscission and dislodgement, and (4) *in situ* measurements of detrital photosynthesis.

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Conflicts of interest

I declare no conflicts of interest.

Data and code availability

Data and annotated code are available at github.com/lukaseamus/plant-limbo. The dataset used for meta-analysis (Wright 2025) can also be accessed at github.com/lukaseamus/detrital-photosynthesis.

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Quantitative Plant Biology

Theories

Plants in limbo — the theory of detrital photosynthesis

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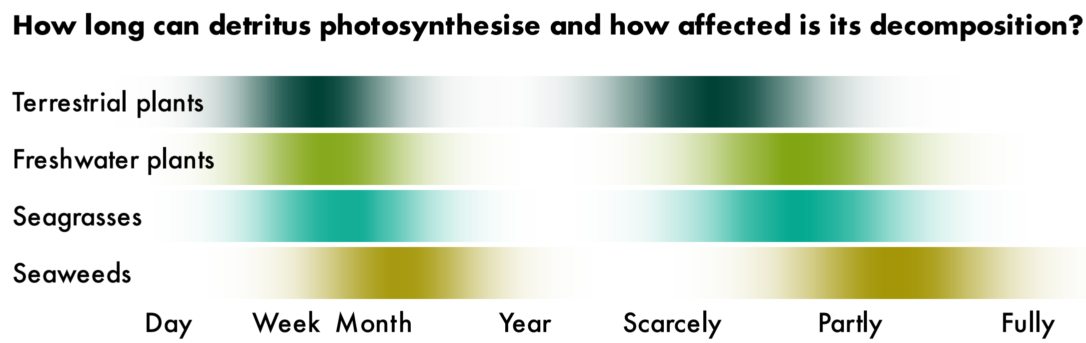
Abstract



Kelp detritus can maintain photosynthesis following detachment. This could be explained by limited tissue differentiation in non-vascular plants, but it remains unclear if detrital photosynthesis is restricted to macroalgae. Here I formalise the theory of detrital photosynthesis by developing a mathematical model and performing a meta-analysis of post-detachment physiology across 129 studies on 92 species of terrestrial and aquatic plants, including original data on the seagrasses *Amphibolis antarctica* and *Halophila ovalis*. Detrital photosynthesis is a general phenomenon, but its longevity in macroalgae is unmatched. Photosynthetic half-life increases from terrestrial plants (11 days) over freshwater Alismatales (14 days) and seagrasses (17 days) to seaweeds (44 days) and only exceeds detrital half-life for seaweeds (3:1). Detrital photosynthesis therefore uniquely determines macroalgal decomposition. This explains why light slows kelp decomposition, contrary to the expected effect of photodegradation. The presented theory questions conventional understanding of decomposition, especially in the context of blue carbon.

Keywords carbon cycle, carbon sequestration, decay, degradation, leaf litter

Introduction

…tossed and swept like dead leaves from one spot to another, never resting, never giving back their goodness to the earth, never fully dead but in some vast limbo between life and extinction, tossing and tumbling without end… — Philip Pullman (1994) *The Tin Princess*

Seaweed detrital dynamics are influenced by physiology. Kelp detritus can remain physiologically viable for months (de Bettignies et al. 2020, Frontier et al. 2021, Wright & Foggo 2021, Frontier et al. 2022, Wright et al. 2022, Wright & Kregting 2023, Wright et al. 2024), often resulting in atypical decomposition trajectories (Frontier et al. 2021, Filbee-Dexter et al. 2022, Frontier et al. 2022, Kennedy & Blain 2025). Floating kelp detritus sinks before it stops photosynthesising (Graiff et al. 2013, 2016, Tala et al. 2019). And if detached tissue happens to be reproductive, it can even be a vector for dispersal (Macaya et al. 2005, Fraser et al. 2018, Tala et al. 2019, de Bettignies et al. 2020, Frontier et al. 2021). This stands in stark contrast to the traditional ecological definition of detritus as "dead organic matter" (Moore et al. 2004, discussed in Wright & Kregting 2023). There is no word for plant material in limbo—detached but alive. Detached seaweed, except for that of holopelagic origin, has always been referred to as detritus. I propose to continue using this term *sensu lato* for any detached plant tissue and detritus *sensu stricto* for detached *and* dead plant tissue. Detrital photosynthesis—defined as the persistence of photosynthesis after detachment (Frontier et al. 2021, Wright et al. 2022, 2024)—provides the strongest evidence in support of the viability of detritus. But the fundamental impact on detrital dynamics only becomes apparent when comparing decomposition rates between live and dead or senescent detritus. Seaweed detritus with impaired physiology clearly decomposes faster (Birch et al. 1983, Brouwer 1996, Hamersley et al. 2015, de Bettignies et al. 2020). This is likely due to maintenance of growth and chemical defence (Tala et al. 2019, de Bettignies et al. 2020, Wright et al. 2022), probably making physiology an important source of detrital recalcitrance (Wright et al. 2022, 2024).

There is currently no baseline for detrital photosynthesis. The term was first introduced quite recently in the context of kelp decomposition (Frontier et al. 2021, Wright et al. 2022, 2024). But there was clearly awareness of the phenomenon among early phycologists who noted the ability of excised tissue to maintain photosynthesis (Drew 1983) and deemed it necessary to pre-kill detritus (Josselyn & Mathieson 1980, Rice & Tenore 1981, Birch et al. 1983, Smith & Foreman 1984, Brouwer 1996). Other fields of botany do not use the term but the amount of research into physiology of excised leaves of terrestrial and freshwater plants is enormous (e.g. Thimann & Satler 1979, Jana & Choudhuri 1980, 1982, Kar & Choudhuri 1986, 1987, Powles et al. 2006, Gauthier & Jacobs 2018, Kar et al. 2021). The literature also includes three studies on seagrasses (Knauer & Ayers 1977, Pellikaan 1982, Strother & Vatta 1986), the only fully marine angiosperms and vascular counterpart to seaweeds (Olsen et al. 2016). Yet no attempt has been made to align these seemingly incongruous research trajectories in a formal comparison. It consequently remains unclear if detrital photosynthesis is unique to seaweeds. It has been suggested that detrital photosynthesis may be explained by the ability of all macroalgal tissues to independently photosynthesise to some extent (Wright & Kregting 2023) but there is no certainty due to a lack of reference points. The mechanism underlying the ability of seaweeds to maintain detrital photosynthesis remains unclear. It could be their aquatic nature which removes reliance on roots for water, nutrient provisioning from the surrounding water, non-vascularity, minimal tissue differentiation and/or something altogether different. To understand the prerequisite for detrital photosynthesis and thus predict prevalence among plants, model systems beyond seaweeds are needed.

Seagrasses are suitable candidates for further investigation of detrital photosynthesis. While also marine macrophytes, seagrasses are fundamentally different from seaweeds. Seagrasses are vascular plants which, due to their terrestrial ancestry (Olsen et al. 2016), are differentiated into roots, rhizomes or stems, and leaves. However, convergent evolution with seaweeds (Olsen et al. 2016) has reduced their reliance on roots for nutrient uptake. Seagrass leaves tend to outperform their roots in suppling nutrients to the plant (Pedersen & Borum 1992, Stapel et al. 1996, Pedersen et al. 1997, Gras et al. 2003, Viana et al. 2019). This suggests that seagrass leaves can independently photosynthesise, seemingly reducing the primary function of the roots to anchorage, analogous to the holdfast of seaweeds. However, given the usually greater ammonium concentration in sediment pore water, roots can supply more nitrogen despite the leaves' greater uptake affinity (Short & McRoy 1984, Lee & Dunton 1999, but see Pedersen & Borum 1992) and nutrients are translocated to leaves (Short & McRoy 1984, Viana et al. 2019), even those tens of centimetres from the source (Marbà et al. 2002). Roots can also tap into nutrient pools that are unavailable to leaves, such as particulate (Evrard et al. 2005) and microbial (Patriquin & Knowles 1972, Mohr et al. 2021) nitrogen. Clearly the photosynthetic emancipation of seagrass leaves is at a halfway point between terrestrial plants and macroalgae, so seagrasses probably display detrital photosynthesis to some extent. Relatively slow and light-dependent chlorophyll degradation in seagrass detritus (Knauer & Ayers 1977, Pellikaan 1982, Strother & Vatta 1986) provides some support for this, but there are currently no data on seagrass detrital photosynthesis.

Here I aim to formalise the theory of detrital photosynthesis in the hope that it will provide a better model for understanding plant decomposition at large. To address this aim, I first conducted a brief *ex situ* decomposition experiment on seagrasses to fill the knowledge gap on seagrass detrital photosynthesis. I chose the paddle weed *Halophila ovalis* (Hydrocharitaceae) and wire weed *Amphibolis antarctica* (Cymodoceaceae) as diverse representatives because of their phylogenetic (diverged 105 Ma ago, Waycott et al. 2018) and morphological (de los Santos et al. 2012, 2016) differences. Senescence of fresh leaves was induced by cutting their petiole and leaving them to decompose on sediment, measuring net oxygen production of destructively sampled leaves at regular intervals over 34 days. On the basis of these and previous (Wright et al. 2022, 2024) observations of detrital photosynthesis, I then developed a mathematical model which describes detrital photosynthesis as a logistic decay function of detrital age. After compiling a detrital photosynthesis dataset (Wright 2025), I used this model in a Bayesian meta-analysis to infer the longevity of detrital photosynthesis for different plants and quantify the effects of light and choice of response variable (photosynthesis vs. chlorophyll). Finally, I compared the persistence of detrital photosynthesis with detrital half-life (Olson 1963) to determine relative physiological influence on decomposition for different plants. This is especially relevant to the case of seaweeds—the current focus of blue carbon (Krause-Jensen & Duarte 2016, Pessarrodona et al. 2023, Filbee-Dexter et al. 2024)—whose carbon cycle is speculated to be strongly influenced by detrital photosynthesis (Wright et al. 2022, 2024).

Methods

Model

Detached plant parts can remain in limbo for long durations, often photosynthesising near baseline levels. For this reason and for simplicity detrital photosynthesis has mostly been described as a linear function of detrital age (Frontier et al. 2021, Wright & Foggo 2021, Wright et al. 2022, Wright & Kregting 2023, Wright et al. 2024). But in two cases a logistic decay function was used: in a beta model predicting chlorophyll fluorescence (Fv/Fm) (Frontier et al. 2022) and in a binomial model predicting the probability of photosynthesis (Wright et al. 2024) with time after excision. The logistic function describes the decay of photosynthesis after detachment better than the linear function because it is bounded. Detrital photosynthesis cannot conceivably exceed baseline photosynthesis under identical conditions. An ill-defined baseline or changes in light, temperature, carbon, nutrients etc. may naturally lead to brief increases after detachment (Frontier et al. 2021, Wright & Kregting 2023), but this cannot be considered part of the underlying process of interest. Similarly, photosynthesis cannot decline beyond a set minimum, which is zero for most response variables (e.g. gross photosynthesis, chlorophyll fluorescence, chlorophyll) and a negative constant for net photosynthesis. Hence the logistic function is the natural choice.

There are various ways to parameterise the logistic function. Typically, a simple linear term with intercept and slope is inserted into the standard logistic function. But for logistic decay with time (t), the intercept is not a meaningful parameter because $f(t = 0)$ is the known baseline. The slope is more meaningful because it provides the logistic rate of decay per unit t, but ultimately the parameter of interest is the sigmoid inflection point or midpoint: t when $f(t) = 0.5$. It has the same units as t and is the only parameter that gives a directly interpretable measure of decay. As such it is analogous to half-life ($t_{½}$), t when $f(t) = 0.5$ for the exponential decay function (Olson 1963). The logistic midpoint can be calculated as intercept / slope, but this usually results in enormous uncertainty and difficulty to calculate meaningful central tendencies (Wright et al. 2024), so the midpoint is best directly estimated. The slope and midpoint parameterisation is sometimes used but these parameters are not identifiable with an intercept near a known baseline. To simplify the model while retaining the midpoint, one must either fix the slope or intercept. Only a constant intercept is sensible and since the baseline is expected near 1, a logistic intercept of 5, translating to $f(0) = 0.99$, is the natural choice.

The most complex case to model is that of net photosynthesis since it has a negative lower bound. Two scaling parameters outside the standard logistic function are required. Thus my full logistic function describing the decay of photosynthesis (p) with time after detachment (t) is

$$p(t) = \frac{\alpha + \tau}{1 + e^{\frac{5(t - \mu)}{\mu}}} - \tau \qquad (1)$$

where α is baseline photosynthesis, μ is the midpoint of photosynthetic decay and τ is baseline respiration. $\tau = 0$ when the lower bound is zero as for gross photosynthesis etc. and $\alpha = 1$ when photosynthesis is normalised relative to the baseline. The naming of parameters is inspired by the word אמת (graecised τμα) which is composed of the first, middle and last letters of the Hebrew alphabet, symbolising life (א), transition (מ) and death (ת). In the story of the golem, the word lends life but removing א changes its meaning from truth to death (מת) and takes life.

Literature review

Following the seagrass experiment (see methods in Supplement), I systematically searched the literature using keyword strings such as "(photosynthesis OR chlorophyll) AND ('induced senescence' OR detrit* OR detach* OR excis* OR cut OR isolate*) AND (week* OR day* OR hour* OR minute* OR time OR age)" in Web of Science (webofscience.com/wos) and Google Scholar (scholar.google.com). Once I had a selection of suitable papers, I used Crossref Metadata Search (search.crossref.org) to find similar papers. I accepted studies on all plants *sensu lato*, the condition being that senescence was induced by excision at an exact time and photosynthesis was repeatedly measured in a timeseries of at least 3 timepoints over any interval to assess viability of the detached tissue. Studies on abscission were not considered since senescence starts at an unknown time prior to detachment. I naturally did not include plants that always live detached, such as the holopelagic seaweeds *Sargassum fluitans* and *S. natans*. I accepted various response variables, including O_2 and CO_2 gas exchange, ^{14}C assimilation, Fv/Fm, photosystem I and II electron transport rates (Mehler and Hill reactions), RuBisCo activity or concentration, carbonic anhydrase activity and photorespiration (glycolate concentration), as well as total chlorophyll,

chlorophyll *a* or a chlorophyll indicator. Microbial photosynthesis was generally assumed to be negligible since most of the mentioned response variables would likely not detect it. Including the present study, this resulted in 129 suitable studies between 1957 and 2025 (see meta-analysis references in Supplement) and 551 experiments (median = 3 per study).

Data were extracted by copying tables or digitising figures using WebPlotDigitizer v4.6 (Rohatgi 2022). In two cases I contacted authors for their raw data and in five cases the data were my own. Raw data were otherwise not available. The resulting dataset (Wright 2025) contains data on 92 species from 23 orders in four non-taxonomic groups of interest: terrestrial plants (66 species), freshwater plants (6 species), seagrasses (5 species) and seaweeds (15 species). It is noteworthy that all freshwater plants in the dataset belong to the order Alismatales, like seagrasses, and all seaweeds belonged to the phyla Chlorophyta and Heterokontophyta (Table S1). The assigned taxonomy follows Plants of the World Online (POWO 2025) and AlgaeBase (Guiry & Guiry 2025). My distinction between freshwater and terrestrial plants is based on whether leaves are submerged and photosynthesis can therefore be considered aquatic. There were no studies on freshwater macroalgae, so seaweeds are assumed to be representative. Several predictor variables including light- and water availability as well as the type of photosynthesis measurement were also recorded. When individual observations could not be extracted, the mean, standard error of the mean (s.e.m.) and sample size (n) were recorded instead. Uncertainty was mostly given as s.e.m., but when standard deviation (s.d.) was provided instead, it was converted as s.e.m. = $\frac{\text{s.d.}}{\sqrt{\text{n}}}$. In a few cases uncertainty was given as a 95% confidence interval (CI) or interquartile range (IQR). In the former case, I converted as s.e.m. = $\frac{0.5 \times \text{CI}_{0.95}}{\Phi^{-1}(0.975)}$, in the latter I assumed mean = median and conservatively converted the larger of the two quartile ranges ($Q_3 - Q_2$ or $Q_2 - Q_1$) as s.e.m. = $\frac{\text{QR}}{\Phi^{-1}(0.75) \times \sqrt{\text{n}}}$.

Meta-analysis

As is typical of meta-analyses, the data contain a mixture of observations and means. To jointly analyse data with at different levels of uncertainty one must either summarise observations and build a measurement error model or expand means to observations by simulation. I opted for the latter and obtained 10646 observations by simulating draws from the sampling distribution. For most response variables, which are bounded by zero, I chose the gamma distribution, for net photosynthesis I chose the normal distribution and for Fv/Fm I chose a truncated normal distribution because some summary statistics violated the beta distribution. I implemented the sampling in R v4.5.2 (R Core Team 2026) using `rgamma( n , mean^2 / sd^2 , mean / sd^2 )`, `rnorm( n , mean , sd )` and `rtnorm( n , mean , sd , 0 , 1 )`, the last function being from extraDistr v1.10.0 (Wolodzko 2023).

Since the response variable is an amalgam of from various measures of photosynthesis, I normalised it to enable formal comparison. Observations were expressed relative to the initial observation or mean of initial observations in the timeseries. The resulting ratios, representing relative photosynthesis, are the least invasive form of normalisation and frequently used to express photosynthetic decay (29 of 129 studies, e.g. Thimann & Satler 1979, Jana & Choudhuri 1980, Pellikaan 1982). The drawback of this approach is that data are not fully brought onto the same scale because net photosynthesis allows zeros and negative ratios. I resolved this issue by replacing all zeros and negative ratios with the minimum observed positive ratio, assumed to be the minimum observable ratio within measurement error of zero. This means equating net respiration with absence of photosynthesis, which is of course not the case because respiration can mask photosynthesis in net photosynthesis measurements. Nonetheless, I believe this to be acceptable for the purposes of determining the timescales of photosynthetic decay. This normalisation allows using the simplest form of Equation 1 with $\tau = 0$ and $\alpha = 1$.

Apart from t (days), the main predictor variables of interest are plant group (terrestrial plants, freshwater plants, seagrasses, seaweeds), light availability and the type of response variable (chlorophyll or photosynthesis). Light is naturally a key predictor of photosynthetic persistence (Frontier et al. 2021, 2022, Wright et al. 2024) and the type of response variable is important because chlorophyll and photosynthesis are expected to senesce on different timescales. Water availability is only relevant to terrestrial plants and was therefore not considered. Since light availability and the type of response variable are binary predictors, they were coded as indicators of light (−0.5 for dark, +0.5 for light) and chlorophyll (−0.5 for photosynthesis, +0.5 for chlorophyll), centred on zero to avoid the bias in prior variance typical of indicator variables (McElreath 2020) and allow easy prediction of the mean. To get a sense of interspecific and experimental variation, I also included plant species (92 levels) and experiment (551 levels) as predictors. The resulting Bayesian log-normal multilevel model of relative photosynthesis (p) is

$$\ln p \sim \mathcal{N}(\bar{p}, \sigma) \tag{2}$$

$$\bar{p} = \ln \frac{1}{1 + e^{\frac{5(t-\mu)}{\mu}}}$$

$$\ln \mu = \alpha_G + \alpha_S + \alpha_E + \beta_G \times l + \gamma_G \times c$$

$$\alpha_G \sim \mathcal{N}(\bar{\alpha}, \sigma_{\alpha_G})$$

$$\alpha_S \sim \mathcal{N}(0, \sigma_{\alpha_S})$$

$$\alpha_E \sim \mathcal{N}(0, \sigma_{\alpha_E})$$

$$\beta_G \sim \mathcal{N}(\bar{\beta}, \sigma_\beta)$$

$$\gamma_G \sim \mathcal{N}(\bar{\gamma}, \sigma_\gamma)$$

$$\bar{\alpha} \sim \mathcal{N}(\ln 14, 0.3)$$

$$\bar{\beta}, \bar{\gamma} \sim \mathcal{N}(0, 0.4)$$

$$\sigma_{\alpha_G}, \sigma_{\alpha_S}, \sigma_{\alpha_E} \sim \mathcal{N}_{0,\infty}(0, 0.3)$$

$$\sigma_\beta, \sigma_\gamma \sim \mathcal{N}_{0,\infty}(0, 0.4)$$

$$\sigma \sim Exp(1)$$

where the logarithm of μ (Equation 1) is described by a linear model with intercepts (α) indexed by plant group (G), species (S) and experiment (E), the effect of light (β) and the effect of measuring chlorophyll relative to photosynthesis (γ) indexed by G. β and γ are differences on the logarithmic scale, so represent log ratios. All parameters are partially pooled to enable prediction for new categories, borrow information and regularise spurious effects (McElreath 2020). The intercept standard deviations (σ_α) give an indication of intergroup, interspecific and experimental variance. The prior for the mean of μ on the logarithmic scale ($\bar{\alpha}$) is centred on ln 14 because two weeks seems like a reasonable median midpoint of photosynthetic decay based on my observations on kelps (Wright et al. 2022, 2024) and literature review. Even though is suspected β and γ to be positive, I centred the priors for $\bar{\beta}$ and $\bar{\gamma}$ on zero to avoid biasing the effect.

The model was written in Stan (Carpenter et al. 2017) and implemented with CmdStan v2.17.1 (Lee et al. 2017) in R. This was facilitated by cmdstanr v0.9.0 (Gabry et al. 2025), tidybayes v3.0.7 (Kay 2024) and bayesplot v1.14.0 (Gabry et al. 2019). Posterior probability distributions for each parameter were estimated based on 8×10^4 Markov chain Monte Carlo samples (10^4 iterations × 8 chains). Model performance was assessed using $\widehat{R}$ scores and trace rank and pair plots (Gabry et al. 2019). For data wrangling, summary and visualisation I used tidyverse v2.0.0 (Wickham et al. 2019), here v1.0.2 (Müller 2025), ggridges v0.5.7 (Wilke 2021), ggh4x v0.3.1 (van den Brand 2025), geomtextpath v0.2.0 (Cameron & van den Brand 2022), patchwork v1.3.2 (Pedersen 2025) and officer v0.7.0 (Gohel et al. 2025). For details, see data and code at github.com/lukaseamus/plant-limbo.

To contextualise μ (days), I formally compared it with the half-life $t_{1/2}$ (days), its equivalent in the exponential model of decomposition. To facilitate the comparison, I hereinafter refer to μ as the photosynthetic half-life and $t_{1/2}$ as the detrital half-life. $t_{1/2}$ is related to the exponential decay constant k by $t_{1/2} = \ln 2 / k$ (Olson 1963). k is usually estimated using the exponential decay function but can also be approximated by the detrital turnover rate, the ratio of detrital production to the detrital pool (Cebrián & Lartigue 2004). Using published data (Cebrián & Lartigue 2004, Anderson-Teixeira et al. 2018), I thus extracted or calculated k for terrestrial plants, freshwater plants, seagrasses and seaweeds, and converted it to $t_{1/2}$. I then modelled $t_{1/2}$ with a log-normal multilevel intercept model and compared its predictions to those for μ from Equation 2. I formalised this comparison as the ratio $\mu / t_{1/2}$, a measure of physiological influence on decomposition. $\mu / t_{1/2} = 1$ implies that detrital photosynthesis reaches half its initial value at the same time as detrital mass. If $\mu / t_{1/2} < 1$, photosynthesis decays faster than mass and is unlikely to influence decomposition but if $\mu / t_{1/2} > 1$, photosynthesis essentially determines decomposition.

Results

Seagrasses

Halophila ovalis photosynthesised for longer post-excision than *Amphibolis antarctica*. After 34 days, the duration of the experiment, all excised *H. ovalis* leaves were still obviously net autotrophic while all excised *A. antarctica* leaves were apparently dead. On a mass basis, *H. ovalis* displayed similar baseline light-saturated net photosynthesis (P_{max}) to *A. antarctica*, but its photosynthetic half-life (μ in Equation 1) was 3.9 ± 1.6 (mean ± standard deviation) times as long (Figure 1, Table 1). On a leaf basis, baseline P_{max} of *A. antarctica* was naturally much greater but again the photosynthetic half-life of *H. ovalis* was 6.8 ± 2.5 times as long (Table 1).

Mean leaf blotted mass of *H. ovalis* (initially 0.052 ± 0.005 g leaf^{-1}) declined at linear rates of 0.58 ± 0.26 mg leaf^{-1} d^{-1} or 1.1 ± 0.43 % d^{-1} while that of *A. antarctica* (initially 0.64 ± 0.053 g leaf^{-1}) only declined at 2 ± 2.7 mg leaf^{-1} d^{-1} or 0.3 ± 0.41 % d^{-1}. Hence it is surprising that excised *H. ovalis* leaves photosynthesised for so much longer. The most likely reason is that periphyton, which I observed colonising *H. ovalis* but not *A. antarctica*, contributed to photosynthesis. This likely exacerbated the observed difference in detrital viability between the two seagrasses. Accounting for these limitations, I expect the average seagrass to have a photosynthetic half-life of more than a week but less than a month.

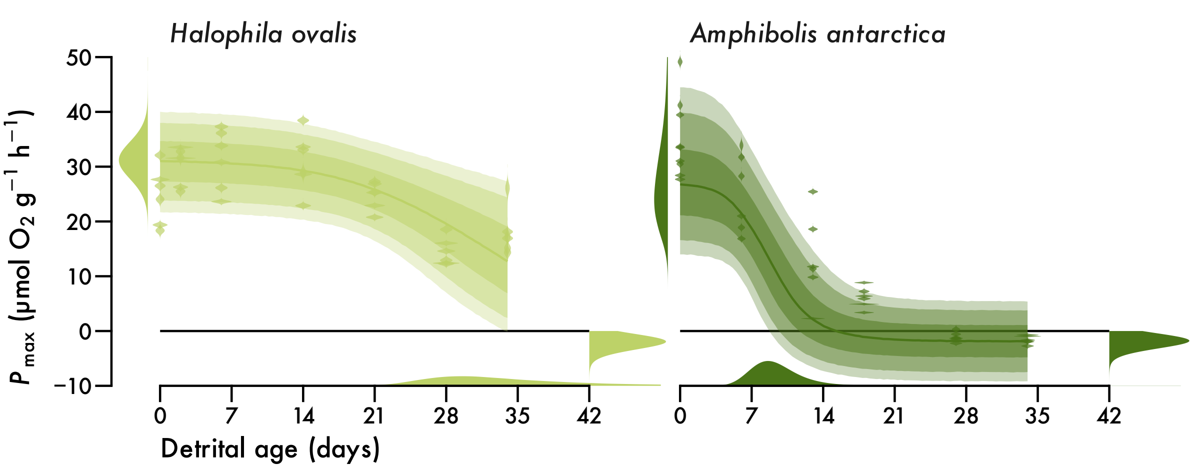


Figure 1. Seagrass detrital photosynthesis. Light-saturated net photosynthesis (P_{max}) of excised *Halophila ovalis* and *Amphibolis antarctica* leaves over 34 days post-excision. P_{max} is standardised by leaf fresh mass. Violins are kernel density estimates of observations with measurement error (n = 42 for *H. ovalis*, n = 39 for *A. antarctica*). Lines and intervals are medians and the central 50%, 80% and 90% of posterior probability for predicted observations. Distributions are posteriors for initial P_{max} (α), photosynthetic half-life (μ) and final respiration (τ) correctly mapped onto the axes (cf. Equation 1).

Table 1. Seagrass parameter estimates (mean ± standard deviation). Δ is the absolute difference between estimates for *H. ovalis* and *A. antarctica*. $P(\Delta > 0)$ is the probability that estimates for *H. ovalis* and *A. antarctica* are different. $P = 0.5$ is equivalent to a coin toss.

Parameter	*Halophila ovalis*	*Amphibolis antarctica*	Δ	$P(\Delta > 0)$
Mass-based				
α (μmol g^{-1} h^{-1})	31 ± 3.4	28 ± 8.4	3.3 ± 9.1	0.69
μ (days)	35 ± 11	9.5 ± 2.6	26 ± 11	1
τ (μmol g^{-1} h^{-1})	2.1 ± 0.85	1.9 ± 0.76	0.17 ± 1.1	0.56
Leaf-based				
α (μmol leaf^{-1} h^{-1})	1.6 ± 0.17	33 ± 12	31 ± 12	1
μ (days)	46 ± 15	7 ± 1.3	39 ± 15	1
τ (μmol leaf^{-1} h^{-1})	0.92 ± 0.38	0.93 ± 0.37	0.0049 ± 0.53	0.51

Meta-analysis

Comparison of detrital photosynthesis across various plants revealed that light ameliorates photosynthetic decay. Detritus of terrestrial plants, seagrasses and seaweeds with access to light is predicted to have a photosynthetic half-life that is 1.8, 1.6 and 2 times as long as that in darkness (Figure 2, Table 2). Freshwater plants form an exception with a mere 68% chance (only somewhat above 50:50) of extended photosynthesis under light (Figure 2, Table 2). The predicted median ratio for new plants is therefore 1.5 (ln ratio = 0.4 ± 0.52). In absolute terms, however, the light effect is only substantial for seaweeds (Figure 2). While median terrestrial photosynthetic half-life is only expected to increase from 7.2 to 13 days with access to light, that of seaweeds increases from 24 to 51 days (Table S2). In darkness, seaweed detritus thus only remains viable for comparable durations to other plants (Figure 2, Table S2). This suggests that the detrital longevity of seaweeds is highly light-dependent.

The choice of response variable also matters when estimating detrital photosynthesis. Chlorophyll generally decays slower than photosynthesis. This pattern is consistent across all plants with photosynthetic half-life predicted to be 1.4 times (ln ratio = 0.33 ± 0.36) as long for chlorophyll as for photosynthesis (Figure 2, Table 2). In absolute terms, the effect again naturally dominates for seaweeds (Figure 2), with an increase from 51 days for photosynthesis to 81 days for chlorophyll under light (Table S2). Measurements based on chlorophyll therefore tend to overestimate the longevity of detrital photosynthesis, especially for seaweeds.

Across light conditions and response variables, the median half-life of detrital photosynthesis ranges from 11 days for terrestrial plants, over 14 and 17 days for freshwater plants and seagrasses to 44 days for seaweeds (Table 2). The prediction for new plants is 17 days (ln days = 2.8 ± 1.1). In relative terms, seaweed detrital photosynthesis persists 3.9, 3.1, and 2.5 times as long as that of terrestrial plants, freshwater plants and seagrasses (Table 3, Figure S2). All vascular plant detritus remains viable for similar durations but there is a tendency for that of freshwater plants to have a somewhat longer photosynthetic half-life compared to terrestrial plants, with that of seagrasses being longer still (Table 3, Figure S2). This suggests that the non-vascular macroalgae have the greatest potential for detrital photosynthesis while among the vascular plants there is increased detrital viability with a shift from a terrestrial to an aquatic and yet again to a marine lifestyle.

Table 2. Meta-analysis parameter estimates (Equation 2) for plant groups. Photosynthetic half-life (μ) is predicted for new experiments on unobserved species across the light (β) and chlorophyll (γ) effects. Exponentiating β and γ yields relative effect sizes (cf. Figure 2). μ is also expressed relative to detrital half-life ($t_{½}$) to quantify the physiological influence on decomposition (cf. Figure 3). All estimates except P are given as median (ln mean ± standard deviation).

Parameter	**Terrestrial plants**	**Freshwater plants**	**Seagrasses**	**Seaweeds**
μ (days)	11 (2.4 ± 0.92)	14 (2.7 ± 0.95)	17 (2.9 ± 0.97)	44 (3.8 ± 0.96)
e^{β}	1.8 (0.61 ± 0.18)	1.2 (0.16 ± 0.37)	1.6 (0.47 ± 0.4)	2 (0.77 ± 0.47)
$P(\beta > 0)$	1	0.68	0.89	0.98
e^{γ}	1.3 (0.29 ± 0.15)	1.5 (0.41 ± 0.26)	1.4 (0.32 ± 0.3)	1.5 (0.47 ± 0.29)
$P(\gamma > 0)$	0.97	0.96	0.89	0.98
$t_{½}$ (days)	409 (6 ± 1.5)	57 (4 ± 1.5)	68 (4.2 ± 1.5)	14 (2.7 ± 1.5)
$\mu/t_{½}$	0.028 (−3.6 ± 1.8)	0.25 (−1.4 ± 1.8)	0.25 (−1.4 ± 1.8)	3 (1.1 ± 1.8)
$P(\mu/t_{½} > 1)$	0.02	0.22	0.23	0.73

Notwithstanding these general trends, there is substantial interspecific and experimental variation in photosynthetic half-life (Figure 3). Median estimates for species range from 3–4 days for the wheat *Triticum aestivum* to 154 days for the Antarctic brown seaweed *Desmarestia anceps* (Table S3). Median estimates for experiments range from 1.2 to 438 days. The variance in photosynthetic half-life between plant groups, species and experiments is directly quantified by logarithmic partial pooling standard deviations (σ_{α} in Equation 2). These inter-group, interspecific and experimental standard deviations are 0.51 ± 0.16, 0.53 ± 0.12 and 0.72 ± 0.057, so experimental variation is the greatest source of uncertainty. Such variation in the physiological viability of detritus may translate to experimental variation in decomposition trajectories.

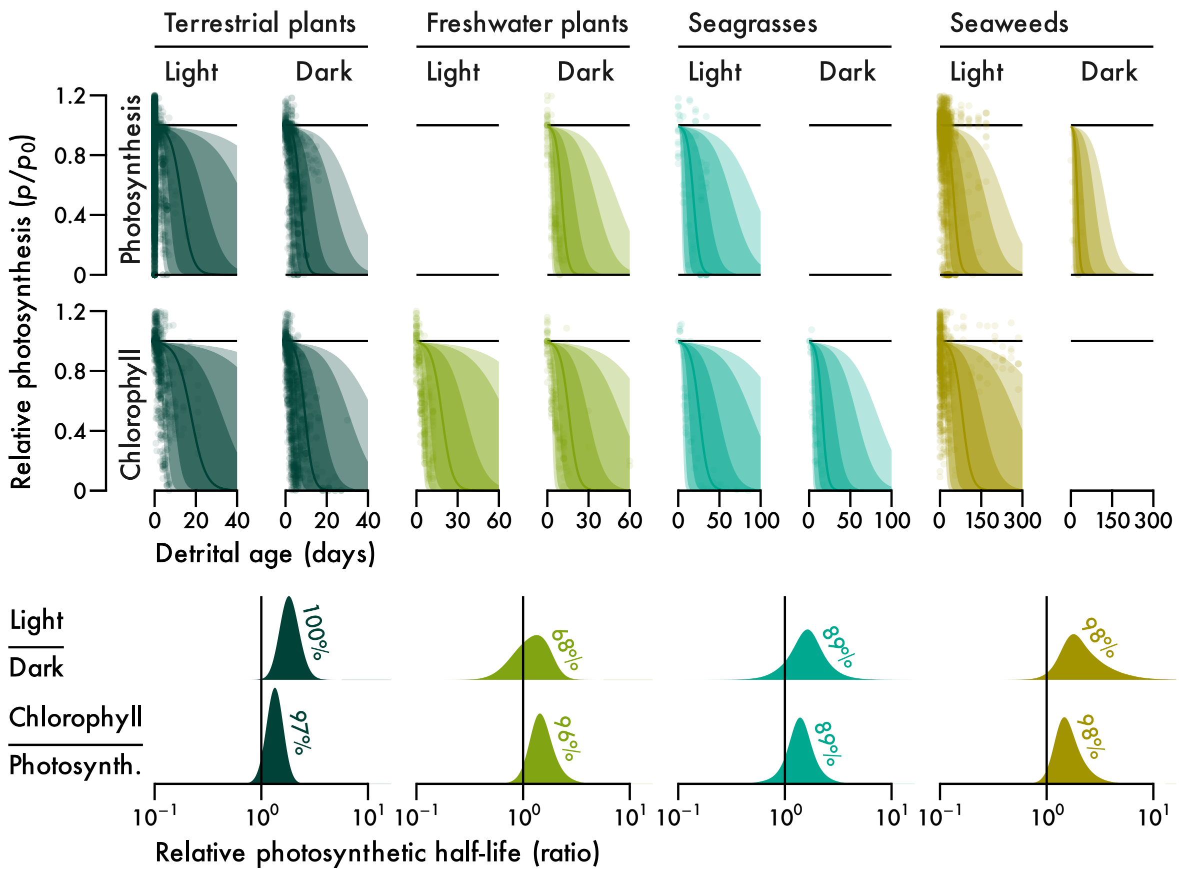


Figure 2. The effect of light and response variable type on detrital photosynthesis for major plant groups: terrestrial plants (Streptophyta excluding aquatic plants), freshwater plants (Alismatales excluding seagrasses), seagrasses, and seaweeds (Chlorophyta and Heterokontophyta excluding freshwater macroalgae). Points are observations (n = 7240, 673, 160 and 2573 for terrestrial plants, freshwater plants, seagrasses and seaweeds). Note that there are several observations outside the axes ranges. Lines and intervals are medians and the central 50%, 80% and 90% of posterior probability for the median prediction ($e^{\bar{p}}$ in Equation 2). Distributions are kernel density estimates of posterior probabilities for the light and chlorophyll relative effect sizes (e^{β} and e^{γ} in Equation 2). Percentages are probabilities for $\beta > 0$ and $\gamma > 0$.

The extent to which detrital photosynthesis can influence decomposition not only depends on photosynthetic half-life because decomposition timescales also vary substantially across plant groups. Terrestrial plant detritus decomposes very slowly, only reaching half its initial mass after over a year on average (Figure 3, Table 2). Aquatic angiosperms decompose much faster, with a median detrital half-life ranging from 57 days for freshwater plants to 68 days for seagrasses (Figure 3, Table 2), only 14% to 17% of terrestrial half-life (Table 3). Seaweeds decompose fastest, with a median detrital half-life of only two weeks (Figure 3, Table 2), 21–26% of aquatic angiosperm half-life and only 3.5% of terrestrial half-life (Table 3). Photosynthetic half-life must therefore be viewed in relation to detrital half-life to assess how affected decomposition is by detrital photosynthesis. The ratio of photosynthetic to detrital half-life quantifies this relative influence, where ratios below 1 indicate little influence and ratios above 1 indicate substantial influence.

Macroalgal decomposition is uniquely affected by physiology. The median ratio of photosynthetic to detrital half-life ranges from 0.028 for terrestrial plants over 0.25 for freshwater plants and seagrasses to 3 for seaweeds (Figure 3, Table 2). Seaweeds are therefore the only plants with a median ratio above 1 and there is a 73% chance that the ratio for any seaweed is above 1, while the probability for aquatic angiosperms is 22–23% and only 2% for terrestrial plants (Figure 3, Table 2). This means that although all plants display detrital photosynthesis to some extent, it does not influence the decomposition of all plants. Terrestrial decomposition is almost certainly not influenced by physiology, while that of aquatic vascular plants is partly affected and macroalgal decomposition is physiologically determined.

Table 3. Photosynthetic (μ) and detrital ($t_{½}$) half-life contrasts between plant groups (cf. Figure S2). P(ratio > 1) is the probability that estimates are different. P = 0.5 is equivalent to a coin toss. All estimates except P are given as median (ln mean ± standard deviation).

Ratio	μ/μ	$P(\mu/\mu > 1)$	$t_{½}/t_{½}$	$P(t_{½}/t_{½} > 1)$
Seaweeds / Terrestrial	3.9 (1.4 ± 1.3)	0.85	0.035 (−3.3 ± 2.1)	0.057
Seaweeds / Freshwater	3.1 (1.1 ± 1.4)	0.80	0.26 (−1.4 ± 2.1)	0.26
Seaweeds / Seagrasses	2.5 (0.93 ± 1.4)	0.75	0.21 (−1.6 ± 2.1)	0.23
Seagrasses / Terrestrial	1.5 (0.43 ± 1.3)	0.63	0.17 (−1.8 ± 2.1)	0.2
Freshwater / Terrestrial	1.3 (0.23 ± 1.3)	0.57	0.14 (−2 ± 2.1)	0.18
Seagrasses / Freshwater	1.2 (0.2 ± 1.3)	0.56	1.2 (0.18 ± 2.2)	0.53

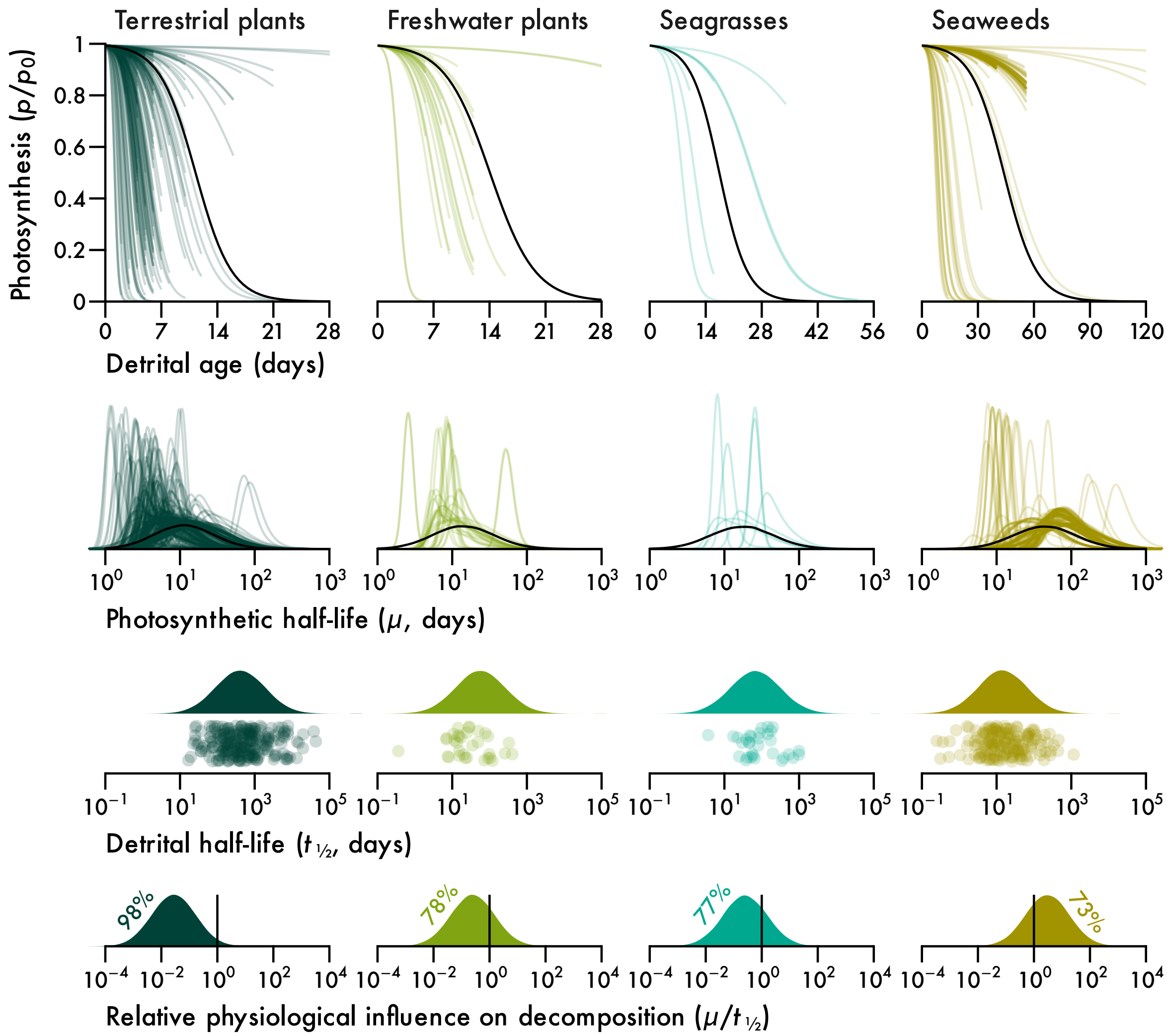


Figure 3. Longevity of detrital photosynthesis relative to detrital mass for major plant groups (cf. Figure 2). Lines are median predictions ($e^{\bar{p}}$ in Equation 2) or posterior probability distributions of photosynthetic half-life (μ in Equation 2). Existing experiments (n = 314, 32, 8 and 197 for terrestrial plants, freshwater plants, seagrasses and seaweeds) are given in colour and predictions for new experiments on unobserved species across the light and chlorophyll effects are given in black. Raincloud plots of detrital half-life ($t_{½}$ *sensu* (Olson 1963) show observations (n = 239, 33, 32 and 171 for terrestrial plants, freshwater plants, seagrasses and seaweeds) below posterior probability distributions for new observations. Distributions in the bottom panel are ratios of the black distributions of μ to the distributions of $t_{½}$. Percentages are probabilities for $\mu/t_{½} < 1$ or $\mu/t_{½} > 1$, depending on what side of the identity line most of the probability lies.

Discussion

Some plant detritus can maintain photosynthesis following detachment. Despite the existence of various relevant literature published on this topic since the 1950s, the phenomenon of detrital photosynthesis had not been explored beyond kelps. In part, this is probably due to diverse research objectives and terminology which obscure the connection between quantitatively comparable data. There was also data scarcity for seagrasses which, as the only vascular counterpart to seaweeds, form an important functional link to freshwater and terrestrial plants. Here I formalised the theory of detrital photosynthesis by collecting the first data on photosynthesis in seagrass detritus, developing a mathematical model of detrital photosynthesis and using it to make the first quantitative comparison of photosynthetic persistence in the detrital phase of various plants. In relation to decomposition timescales estimated by exponential decay, the longevity of detrital photosynthesis provides insight into physiological influence on the fate of detritus. I predict that detrital photosynthesis is a general phenomenon but its effect on decomposition ranges from negligible for terrestrial plants over measurable for aquatic vascular plants to substantial for macroalgae. In fact, it is likely that physiology determines macroalgal decomposition, which would make the initial physiological state of detritus and its light environment the most important predictors of decomposition.

Detrital photosynthesis *per se* is not restricted to non-vascular plants. Seagrass detritus can also photosynthesise for at least a couple weeks. The physiology of diverse seagrasses responds similarly to detrital age and observed differences may be attributable to varying colonisation by periphyton which, given a leaf mass per area of 2.7 and 6.8 mg cm^{-2} for *Halophila ovalis* and *Amphibolis antarctica* respectively (de los Santos et al. 2012, 2016), could contribute 12–172% of initial seagrass photosynthesis (Neely & Wetzel 1997). Alternatively, *A. antarctica* leaves may be more reliant on their roots for nutrient provisioning. But there is no evidence to suggest this (Pedersen et al. 1997) and the 26–39-day shorter photosynthetic half-life of isolated *A. antarctica* leaves compared to *H. ovalis* is surprising given their 1.7 times ($\Delta = 47$ days) longer attached lifespan (Hemminga et al. 1999). Seagrass detrital photosynthesis, despite persisting for weeks, may not necessarily counteract decomposition. Seagrass detritus has a half-life of 68 days on average and complete decomposition may take a year (Harrison 1989, Cebrián et al. 1997, Cebrián & Lartigue 2004, Trevathan-Tackett et al. 2020). While live *Halophila decipiens* ramets do indeed decompose slower than buried ones (Kenworthy et al. 1989), decomposition rates are similar between live and senescent or buried *Thalassia testudinum* leaves (Rublee & Roman 1982, Fourqurean & Schrlau 2003, Rosch & Koch 2009) and reviews suggest that live seagrass leaves actually tend to decompose faster than senescent or dead ones (Harrison 1989, Trevathan-Tackett et al. 2020).

Macroalgae uniquely exhibit long-term detrital photosynthesis, with extreme cases persisting up to a year. As expected from their aquatic lifestyle, seagrasses and freshwater plants exhibit intermediate detrital longevity between seaweeds and terrestrial plants, whose leaves photosynthesise for less than two weeks following excision. The extreme contrast to terrestrial plants has been demonstrated but is in fact somewhat dampened by my formal analysis. With access to light but not water, which is the normal fate of terrestrial plant detritus, detached leaves of various trees tend to lose the ability to photosynthesise in just a few minutes (e.g. (Gauthier & Jacobs 2018, Kar et al. 2021) due to breakdown of stomatal conductance following a loss in water pressure (Powles et al. 2006, Gauthier & Jacobs 2018). Surprisingly seagrasses detritus tends to photosynthesise for longer than that of freshwater Alismatales. The adoption of a marine lifestyle and convergent evolution with seaweeds (Olsen et al. 2016) may be the ultimate reason. But a more proximate explanation is excess production of hydrogen peroxide in detached freshwater plant leaves (Jana & Choudhuri 1982, Kar & Choudhuri 1986, 1987). As demonstrated for *Hydrilla verticillata* (Kar & Choudhuri 1986, 1987), hydrogen peroxide production is exacerbated by photosynthesis, leading to light-induced chlorophyll degradation that is independent of photodegradation. Hence freshwater plants were the only group for which I did not find a positive light effect on photosynthetic half-life. In contrast, there is no evidence for increased chlorophyll breakdown in isolated seagrass leaves exposed to light (Strother & Vatta 1986) with photodegradation only being apparent in dead leaves (Vähätalo et al. 1998). Perhaps the answer lies in alterations to the light harvesting complexes and cell wall features of seagrass leaves that enhance gas exchange (Olsen et al. 2016). Regardless, emancipation of leaf photosynthesis from the rest of the plant is apparently favoured by an aquatic but even more so by a marine lifestyle.

Macroalgal decomposition is uniquely determined by physiology. The influence of detrital photosynthesis on decomposition depends on its longevity relative to decomposition timescales. Of all considered plants, terrestrial ones decompose slowest and macroalgae decompose fastest, so the observed divergence in photosynthetic decay is increased when viewing it relative to decomposition. Macroalgae are the only plants whose detritus photosynthesises for as long as it remains intact, aquatic vascular plants maintain photosynthesis for a quarter of their detrital phase and terrestrial decomposition is unaffected by physiology due to a mismatch between rapid photosynthetic decay and slow decomposition. Since light increases photosynthetic half-life, detrital photosynthesis provides a mechanism contrary to photodegradation by which light slows macroalgal decomposition. Whether these predictions based on excision-induced senescence are realistic depends on the initial physiological state of detritus. Seagrass detritus often incudes green leaves (Knauer & Ayers 1977, Ochieng & Erftemeijer 1999, Jiménez-Ramos et al. 2023). This may in part be due to low nutrient resorption prior to abscission relative to terrestrial plants (Harrison 1989, Pedersen & Borum 1992, Pedersen et al. 1997, Hemminga et al. 1999, Rosch & Koch 2009). But the most likely mechanism is premature detachment caused by external factors such as hydrodynamics and herbivory (Cebrián et al. 1997) paired with seasonal weakening of the ligule (Jiménez-Ramos et al. 2023). Seaweed detritus includes whole plants which are familiar as stranded wrack but many of which are not cast up on the beach to die (van der Mheen et al. 2024, Simpkins et al. 2025). Dislodged plants account for 26% (median, range = 5.6–74%) of kelp detrital production (Gerard 1976, Newell et al. 1982, de Bettignies et al. 2013, Pessarrodona et al. 2018, Pedersen et al. 2020, Gilson et al. 2023) and are not expected to exhibit impaired photosynthesis like distally eroded or abscised tissues (Küppers & Kremer 1978, Arnold & Manley 1985, Rodriguez et al. 2016). Some viable seagrass and seaweed detritus is clearly produced in nature and will therefore follow the predictions of my model.

The theory of detrital photosynthesis is at odds with conventional understanding of the macroalgal carbon cycle. Seaweed blue carbon has been proposed as a substantial contributor to marine carbon sequestration (Krause-Jensen & Duarte 2016, Pessarrodona et al. 2023, Filbee-Dexter et al. 2024). Each proposed sequestration pathway (Pessarrodona et al. 2023) naturally hinges on decomposition. Even sequestration estimates based on deep ocean retention which does not discriminate between organic and inorganic carbon (Filbee-Dexter et al. 2024) are affected by decomposition (White & Norkko 2025). I have previously suggested that detrital photosynthesis is an important determinant of kelp decomposition (Wright et al. 2022, 2024) and have here shown this effect to be unique to macroalgae. The key implication is that seaweed decomposition is not comparable to that of other plants. Due to enormous physiological variation in seaweed detritus, as shown here, it is even hard to compare decomposition within seaweeds when not explicitly accounting for detrital photosynthesis (Kennedy & Blain 2025, White & Norkko 2025). Most of our understanding of seaweed decomposition is based on freshly excised blades (Frontier et al. 2021, Filbee-Dexter et al. 2022, Frontier et al. 2022, Wright et al. 2022, Kennedy & Blain 2025, White & Norkko 2025) as opposed to senescent fragments (Hamersley et al. 2015, de Bettignies et al. 2020) or dead tissue (Josselyn & Mathieson 1980, Rice & Tenore 1981, Birch et al. 1983, Smith & Foreman 1984, Brouwer 1996). But as mentioned, only 26% of seaweed detritus can be considered fully viable when exported. Additionally, decomposition experiments are usually carried out in mesocosms or the shallow subtidal with access to light, but seaweed carbon sinks are said to be found in the darkness of the deep sea (Krause-Jensen & Duarte 2016, Filbee-Dexter et al. 2024, van der Mheen et al. 2024, Simpkins et al. 2025). My meta-analysis suggests that darkness has a detrimental effect on the detrital photosynthesis of seaweeds, thus accelerating their decomposition. Perhaps it is time to question existing preconceptions in macroalgal blue carbon.

To conclude, I show that all plants may photosynthesise following detachment, but timescales vary greatly from a few minutes in tree leaves without access to water to a year for certain seaweeds. Only macroalgal detrital photosynthesis decays on similar timescales to detrital mass. Physiology is therefore expected to determine the decomposition of macroalgae, a relationship which is unique among plants. Predictors of photosynthesis, such as temperature and light, can hence be used to predict macroalgal decomposition. With the theory of detrital photosynthesis formalised, we now need (1) a model of macroalgal decomposition that takes into account this theory, (2) seagrass and seaweed decomposition experiments comparing live and dead detritus to directly investigate the proposed effect of physiology on decomposition (cf. Birch et al. 1983, Brouwer 1996, Hamersley et al. 2015, de Bettignies et al. 2020), (3) studies on the initial physiological state of detritus produced by erosion, abscission and dislodgement, and (4) *in situ* measurements of detrital photosynthesis.

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Conflicts of interest

I declare no conflicts of interest.

Data and code availability

Data and annotated code are available at github.com/lukaseamus/plant-limbo. The dataset used for meta-analysis (Wright 2025) can also be accessed at github.com/lukaseamus/detrital-photosynthesis.

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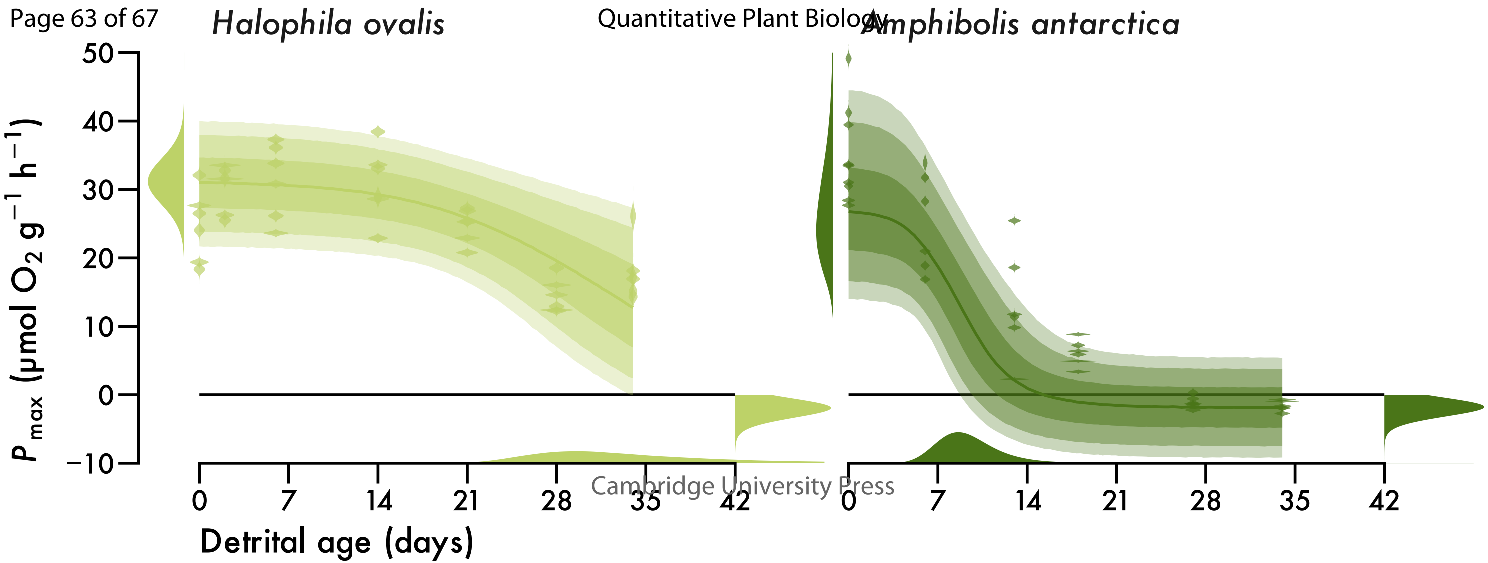
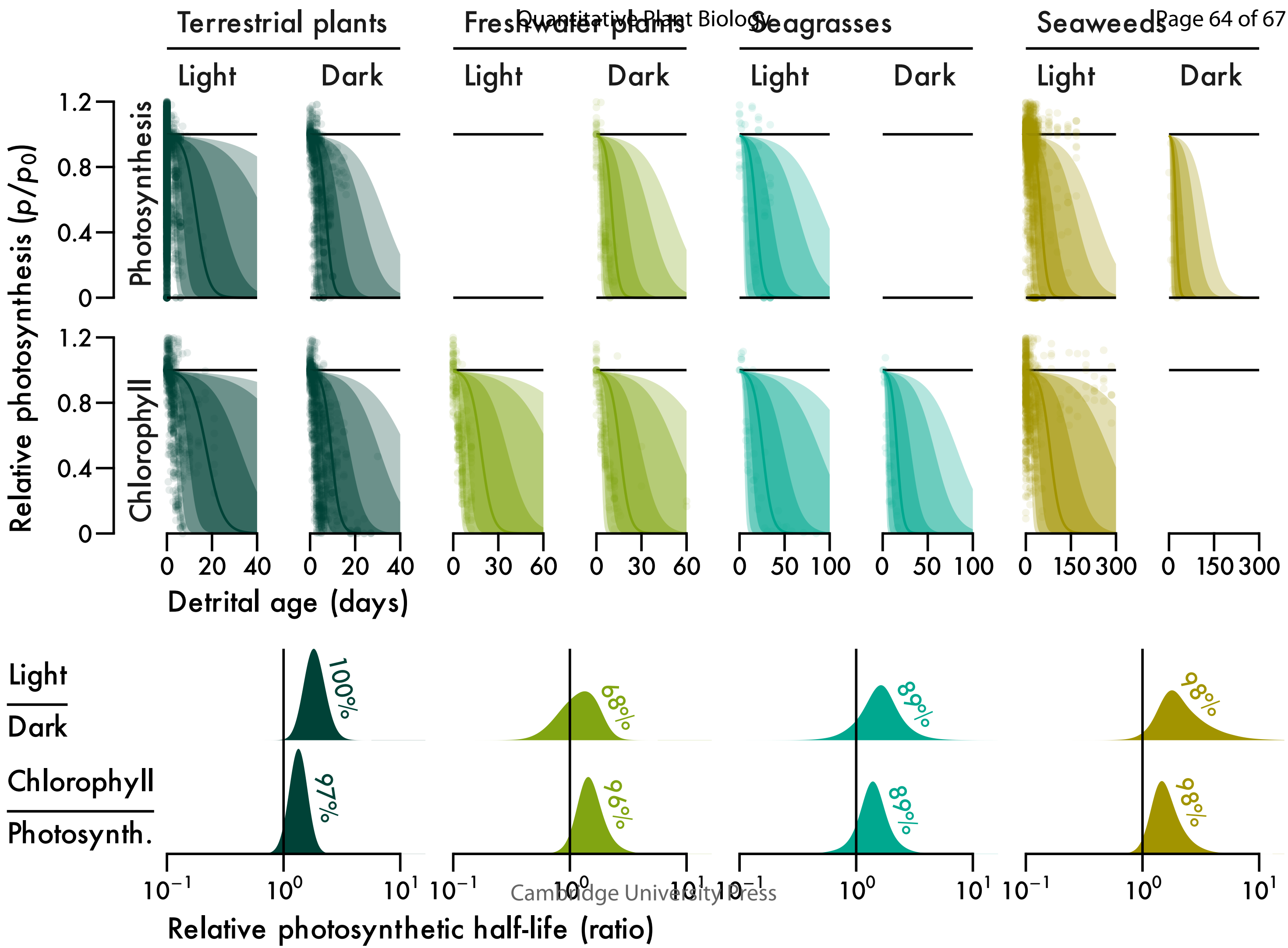
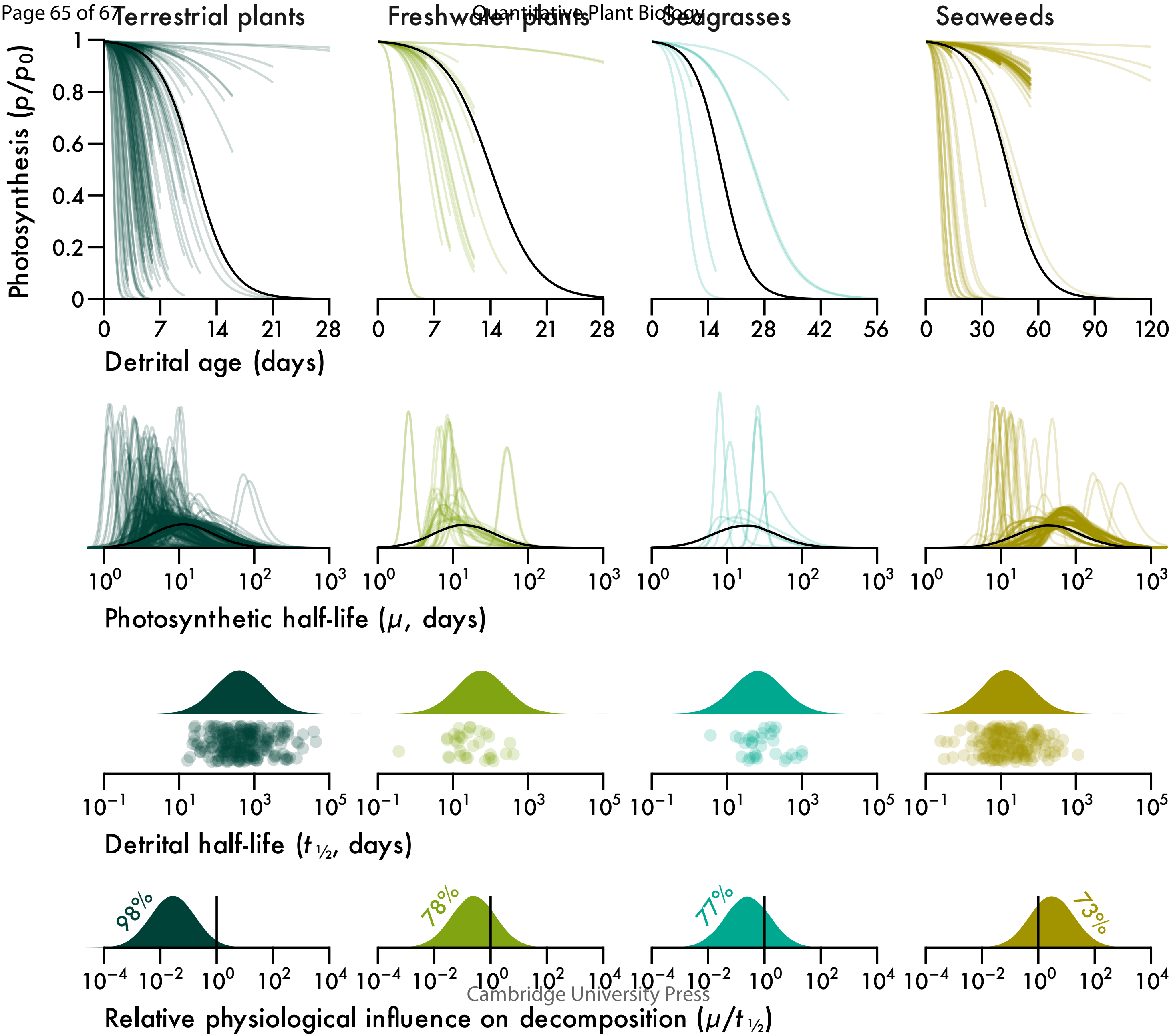
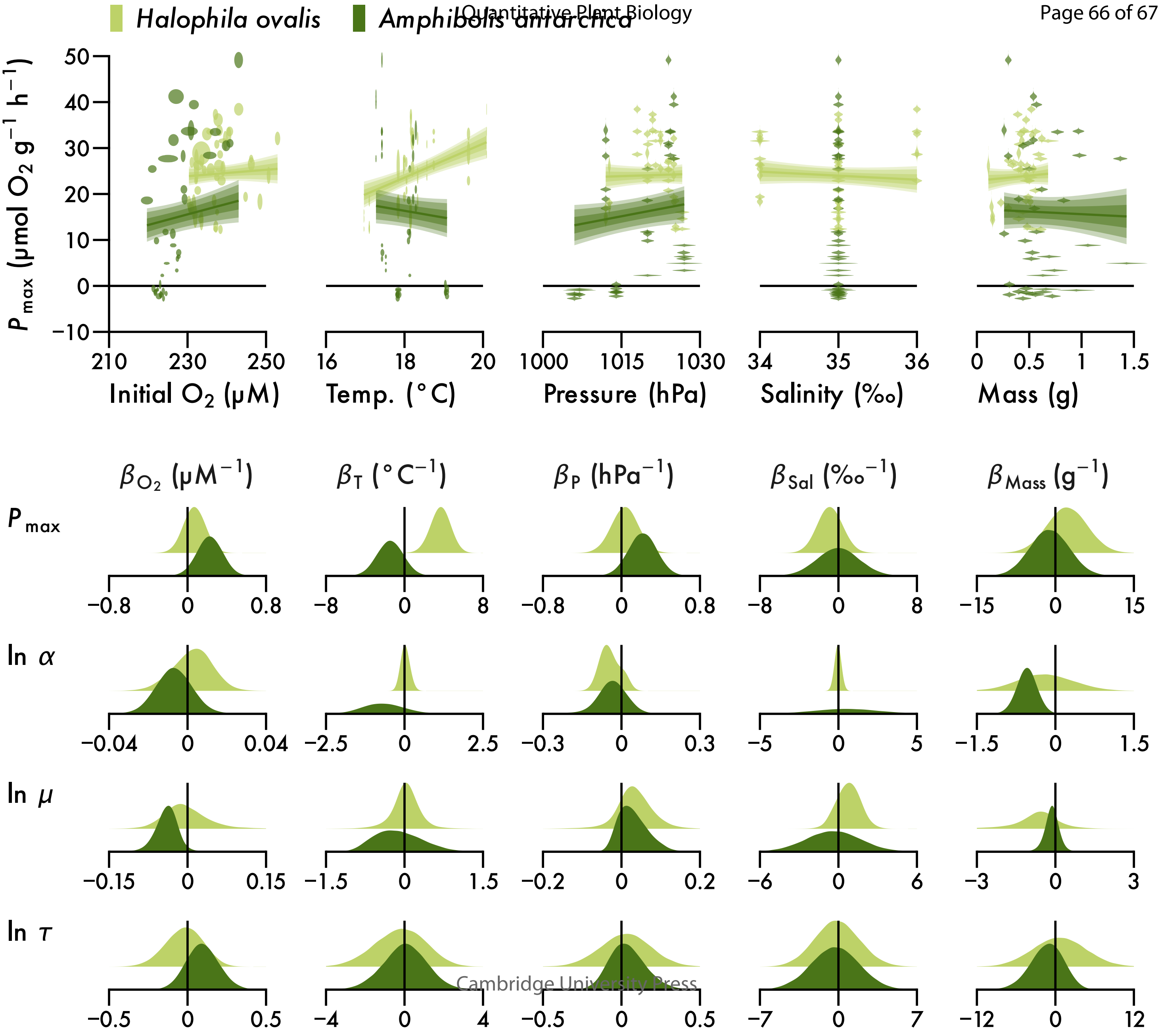
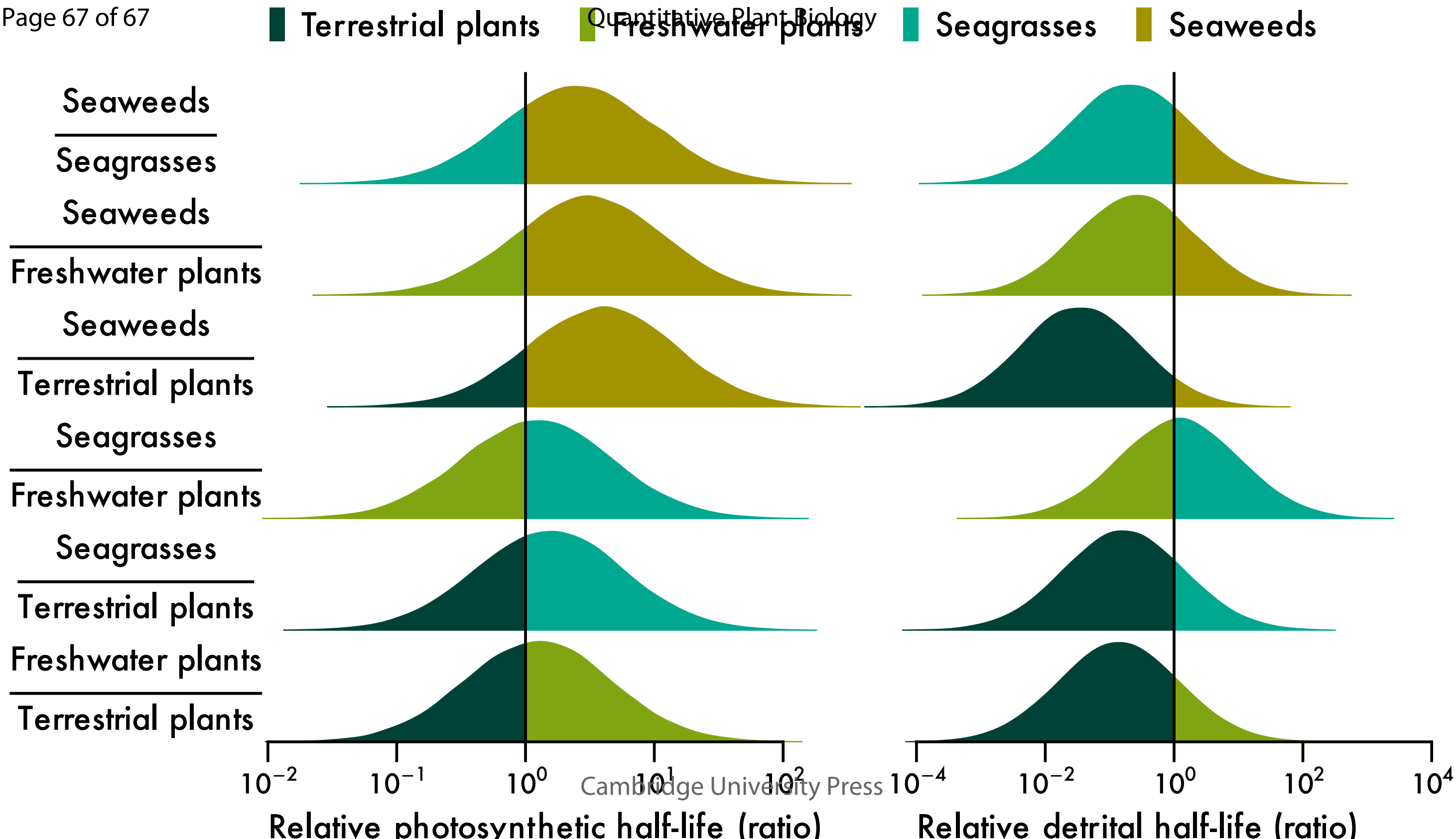

Halophila ovalis
Amphibolis antarctica
P_{max} (μmol O_2 g^{-1} h^{-1})
50
40
30
20
10
0
−10
0
7
14
21
28
35
42
Detrital age (days)


Terrestrial plants
Freshwater plants
Seagrasses
Seaweeds
Light
Dark
Photosynthesis
Chlorophyll
Relative photosynthesis (p/p_0)
1.2
0.8
0.4
0
0 20 40
0 30 60
0 50 100
0 150 300
Detrital age (days)
Light
Dark
Chlorophyll
Photosynth.
100%
89%
89%
98%
97%
96%
89%
98%
10^{-1}
10^{0}
10^{1}
Relative photosynthetic half-life (ratio)


Terrestrial plants
Freshwater plants
Seagrasses
Seaweeds
Photosynthesis (p/p0)
1
0.8
0.6
0.4
0.2
0
0 7 14 21 28
0 7 14 21 28
0 14 28 42 56
0 30 60 90 120
Detrital age (days)
10^0 10^1 10^2 10^3
Photosynthetic half-life (μ, days)
10^−1 10^1 10^3 10^5
Detrital half-life (t½, days)
98%
78%
77%
73%
10^−4 10^−2 10^0 10^2 10^4
Relative physiological influence on decomposition (μ/t½)


Halophila ovalis
Amphibolis antarctica
P_{max} (μmol O_2 g^{-1} h^{-1})
Initial O_2 (μM)
Temp. (°C)
Pressure (hPa)
Salinity (‰)
Mass (g)
β_{O_2} (μM^{-1})
β_T (°C^{-1})
β_P (hPa^{-1})
β_{Sal} (‰$^{-1}$)
β_{Mass} (g^{-1})
P_{max}
ln α
ln μ
ln τ


Terrestrial plants
Freshwater plants
Seagrasses
Seaweeds
Seaweeds
Seagrasses
Seaweeds
Freshwater plants
Seaweeds
Terrestrial plants
Seagrasses
Freshwater plants
Seagrasses
Terrestrial plants
Freshwater plants
Terrestrial plants
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Relative photosynthetic half-life (ratio)
10^{-4}
10^{-2}
10^{0}
10^{2}
10^{4}
Relative detrital half-life (ratio)